

Fast Geometric Machine Learning Methods for Brain Image Analysis

To enhance and streamline cortical surface processing

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Outline

Background

Literature Review

Presentation of CorticalFlow⁺⁺

Insights into Spherical Demons

Presentation of NC-Reg

Conclusion



Background

Literature Review

Presentation of CorticalFlow⁺⁺

Insights into Spherical Demons

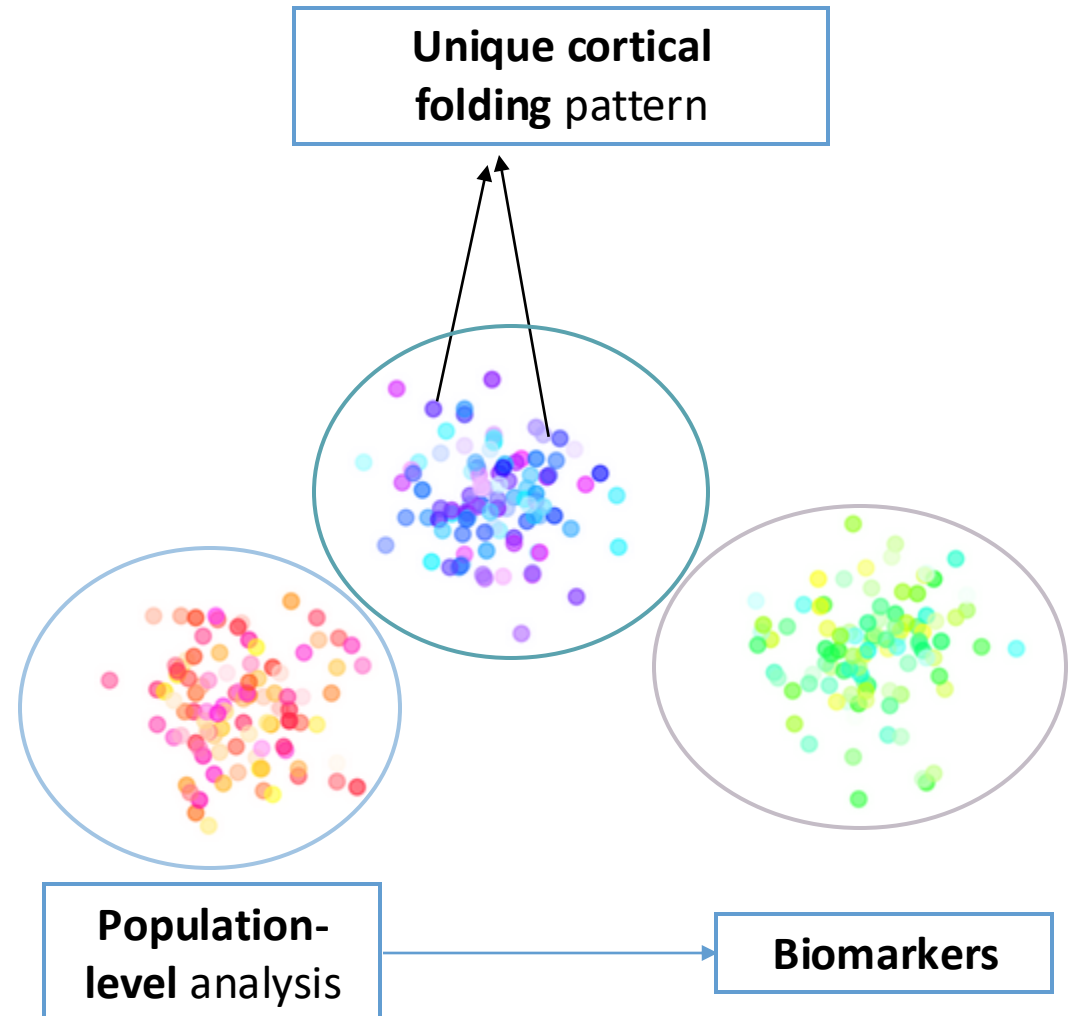
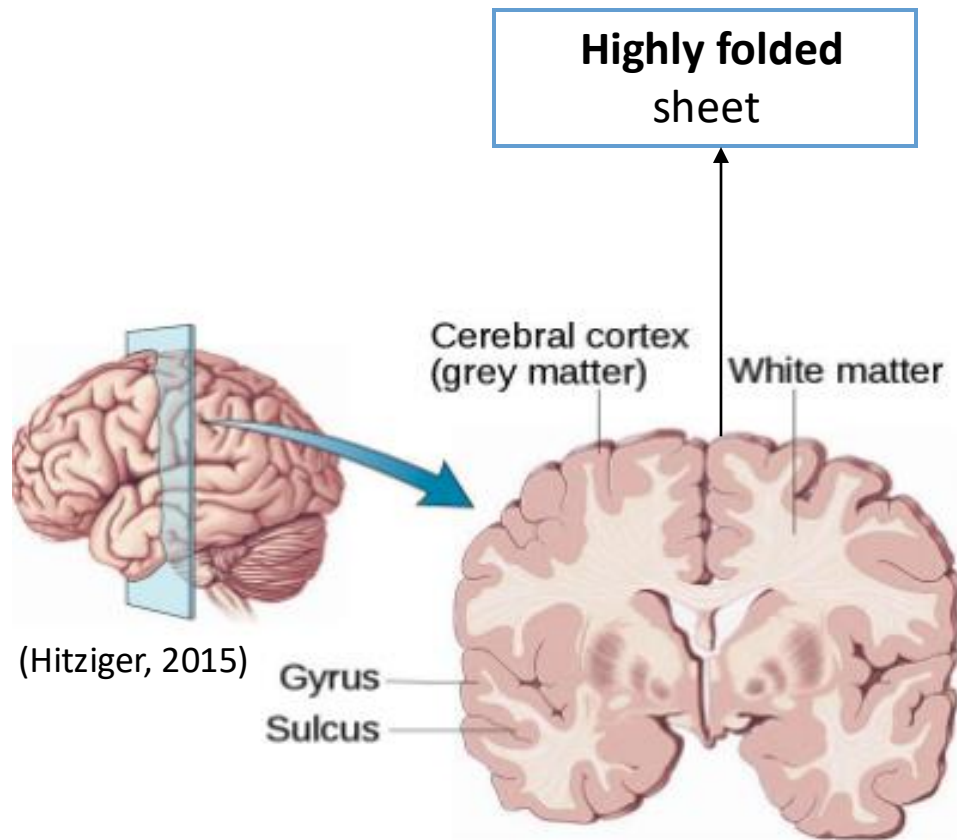
Presentation of NC-Reg

Conclusion

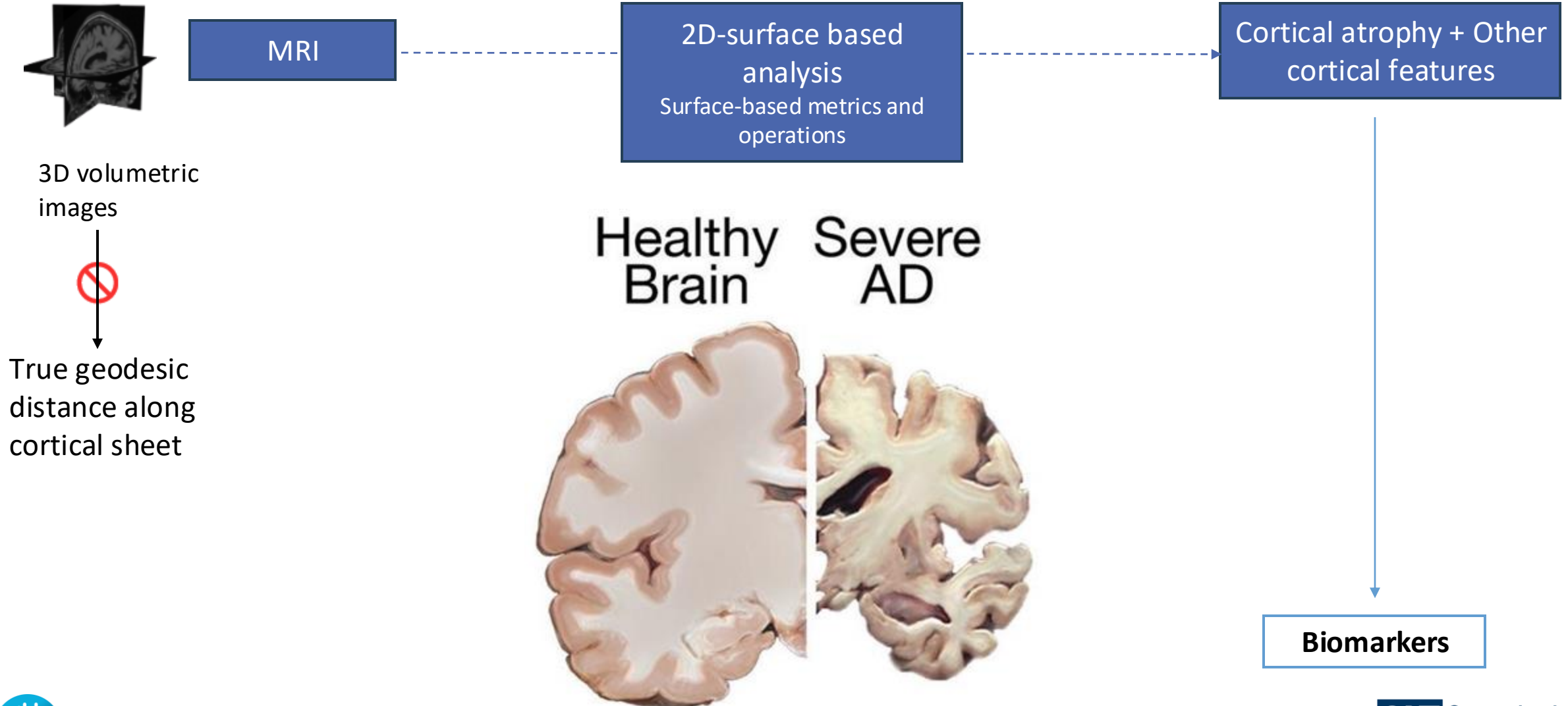
Why population-based
analyses are important
and challenging ?



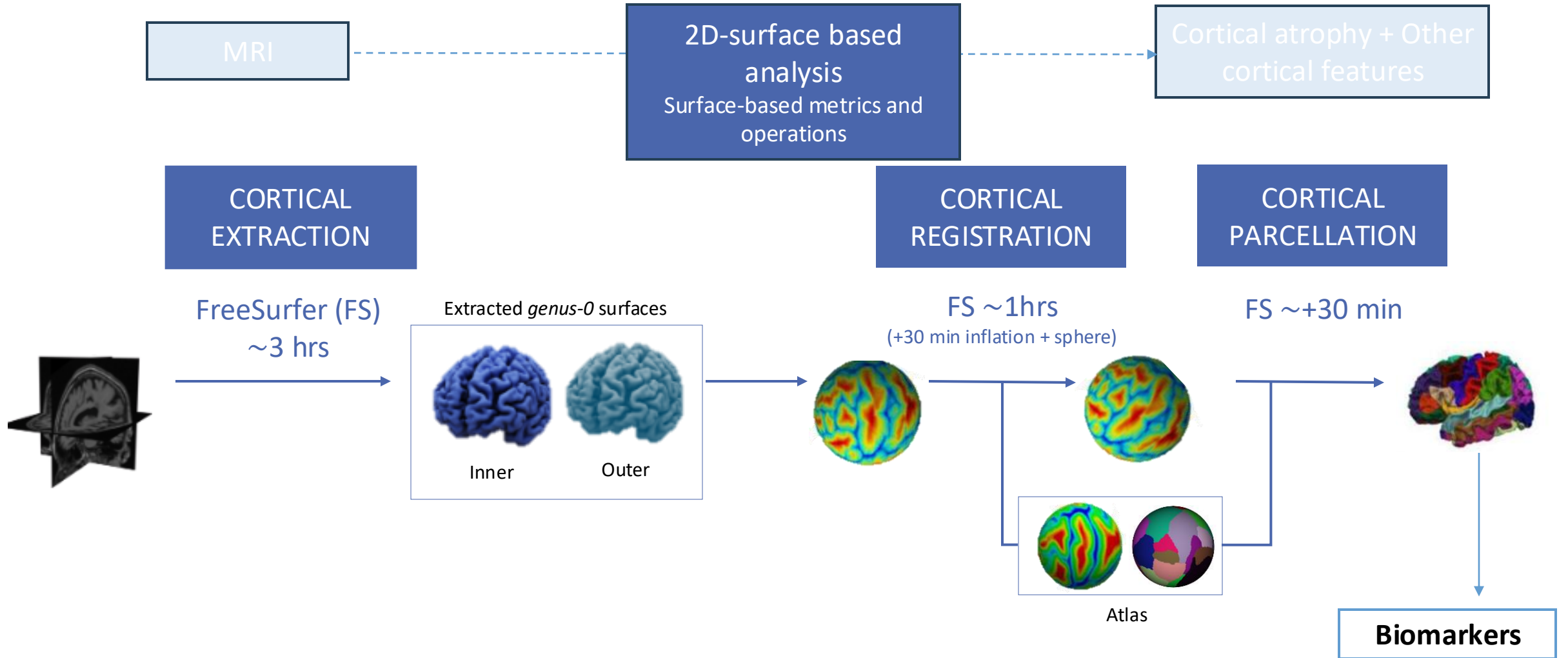
Background

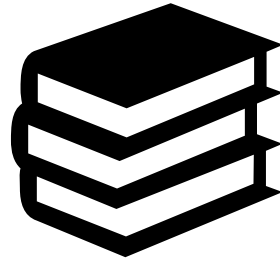


2D-surface more suitable than 3D



2D-surface based processing





Literature Review

Cortical Mesh Reconstruction
Cortical Surface Registration
Neural Representation

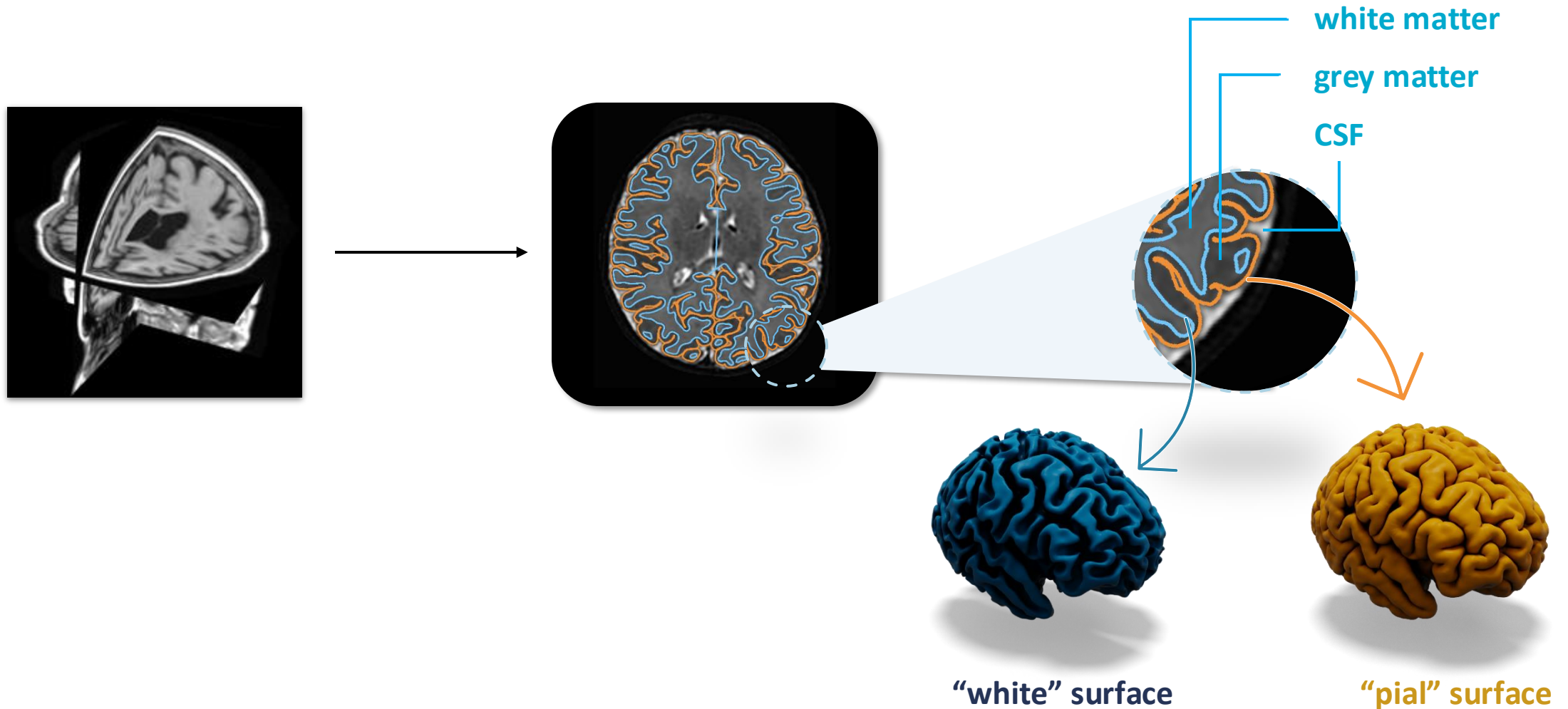
Presentation of CorticalFlow⁺⁺

Insights into Spherical Demons

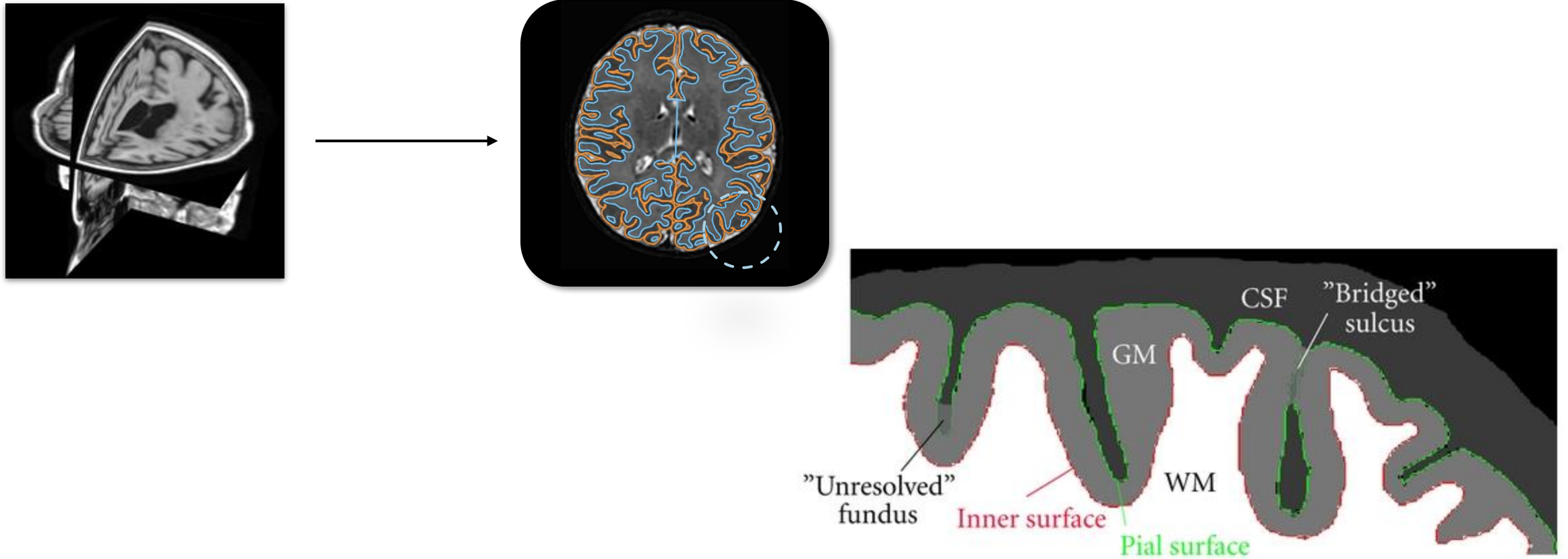
Presentation of NC-Reg

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Cortical Surface Reconstruction



Cortical Surface Reconstruction

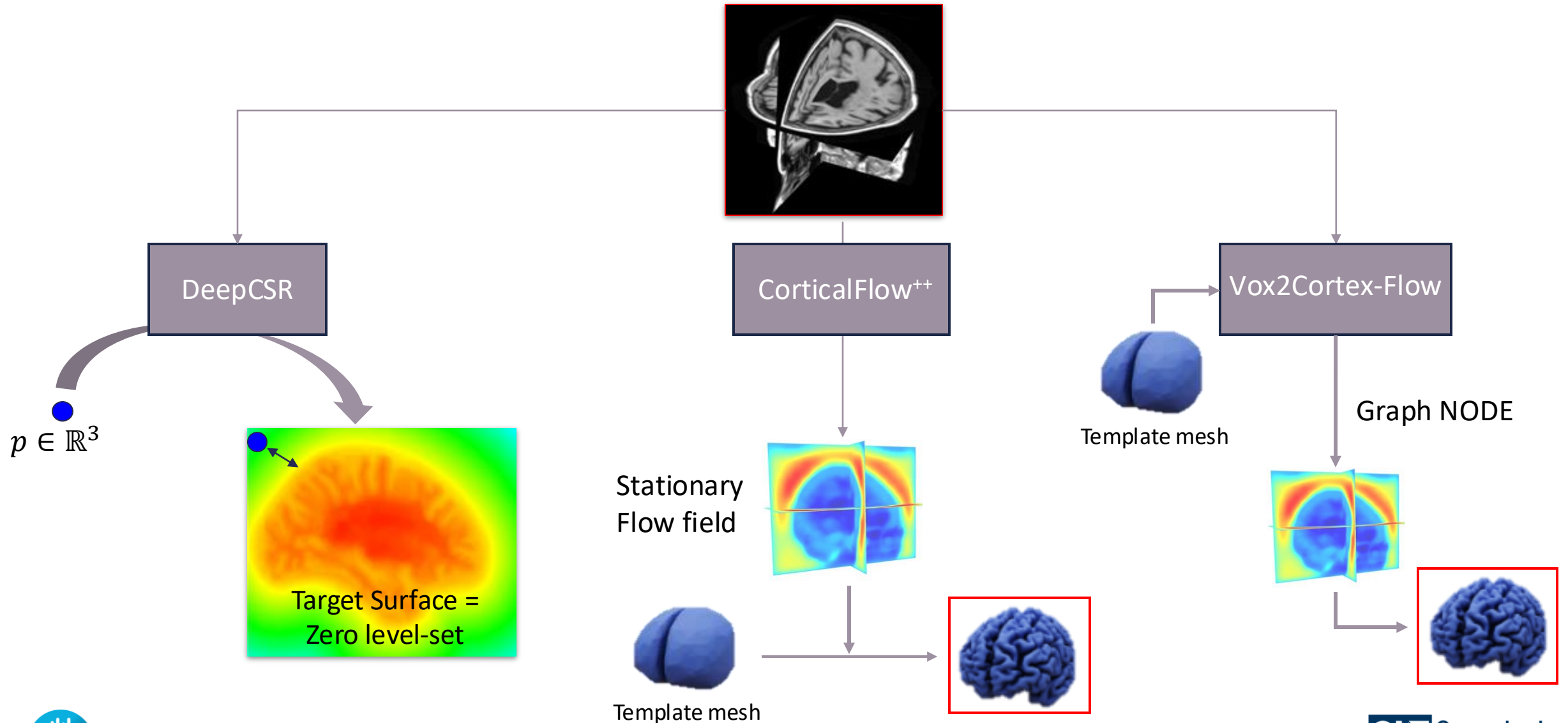


Cortical Surface Reconstruction

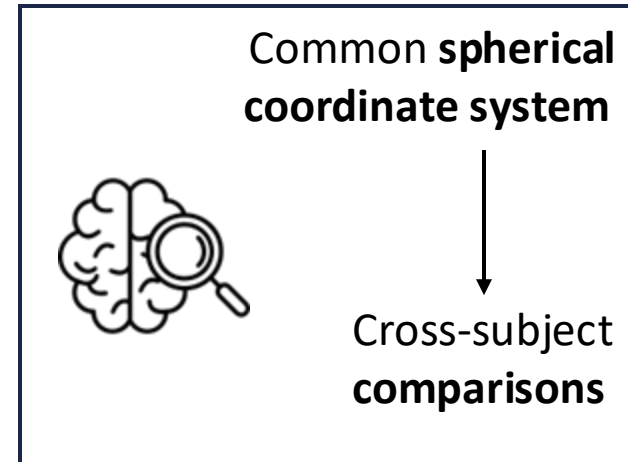
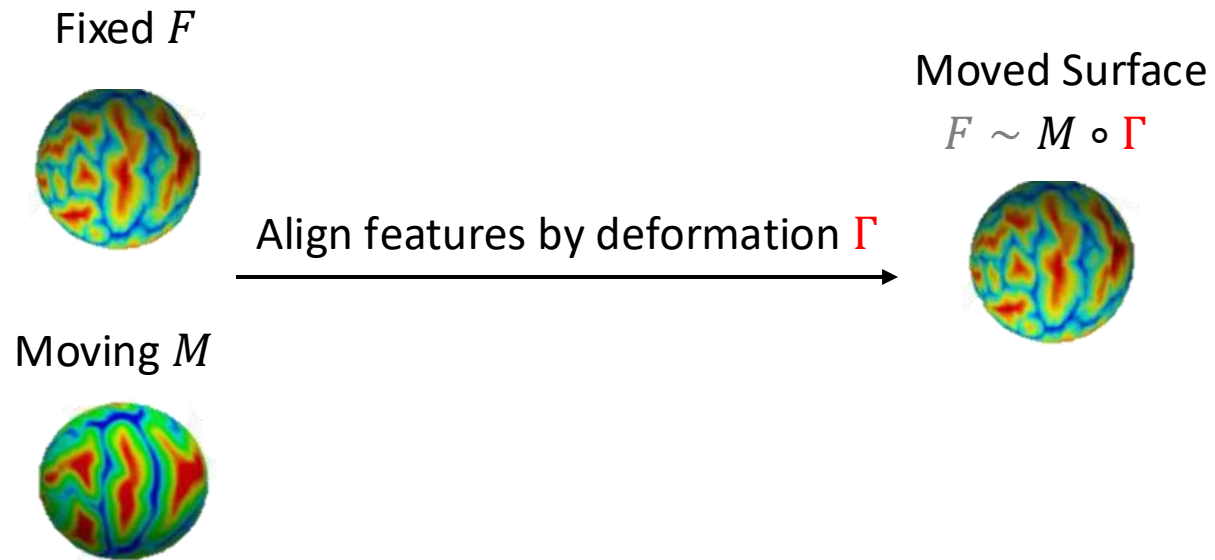


FreeSurfer V6 comprises at least **23 imaging processing methods** and takes more than **6 hours per scan**

Cortical Surface Reconstruction



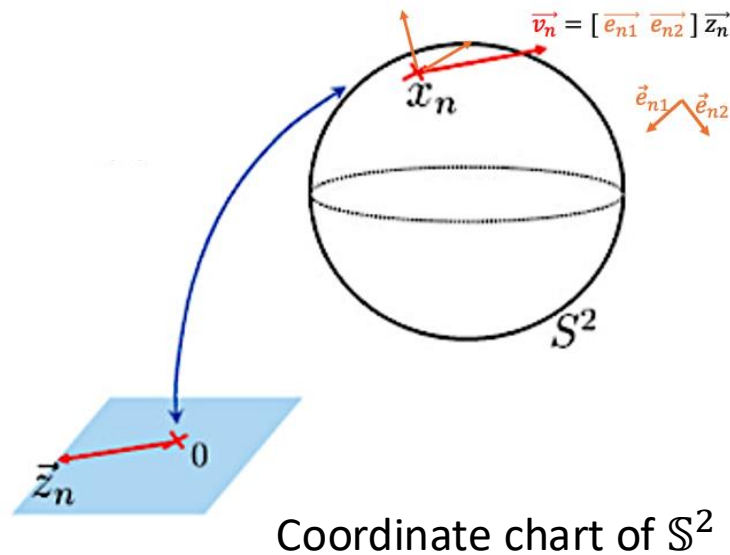
Spherical Surface Registration



$$\arg \min_{\Gamma} \text{Sim}(F, M \circ \Gamma) + \mathcal{R}(\Gamma)$$

Spherical Surface Registration

Spherical Demons (SD)
(Yeo et al., 2010)



Extends

Diffeomorphic Demons
for image registration

to

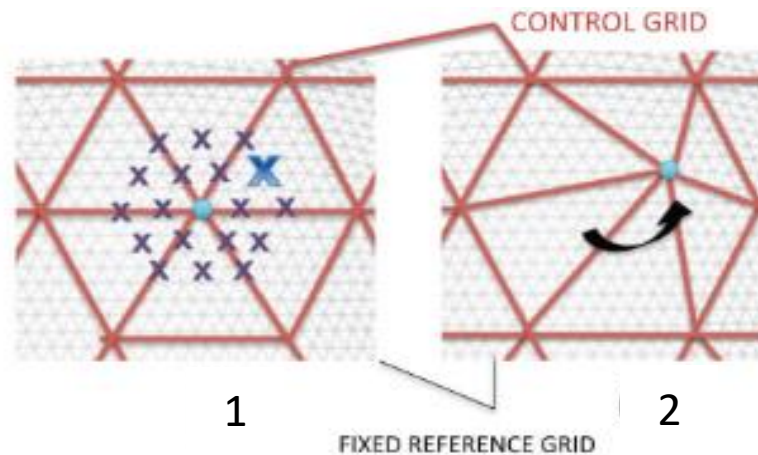
the **unit sphere**

Spherical Surface Registration

Spherical Demons (SD)
(Yeo et al., 2010)

Discrete Markov Random Field (MRF)

- ✓ Reduces the deformation search space
- ✓ Avoids being stuck in local minima
- ⚠ Still computationally intensive for big meshes



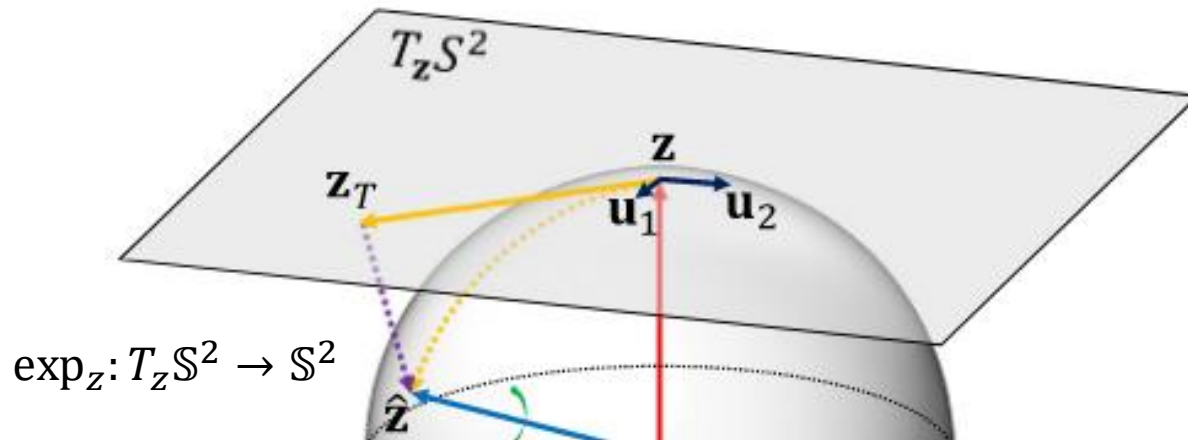
Non-Deep Learning Methods

Spherical Demons (SD)
(Yeo et al., 2010)

Spherical Surface Registration

Multimodal Surface
Matching (MSM)
(Robinson et al., 2014)

Local rotation parameters → Spherical harmonics



- ✓ Template-free groupwise registration
- ✓ Reduced distortions
- ⚠ Fine-tuning of degree of spherical harmonic basis
- ⚠ Still long even with CUDA implementation

Non-Deep Learning Methods

Spherical Demons (SD)
(Yeo et al., 2010)

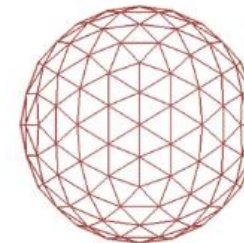
Multimodal Surface
Matching (MSM)
(Robinson et al., 2014)

Spherical Surface Registration

HSD
(Lyu et al., 2019)

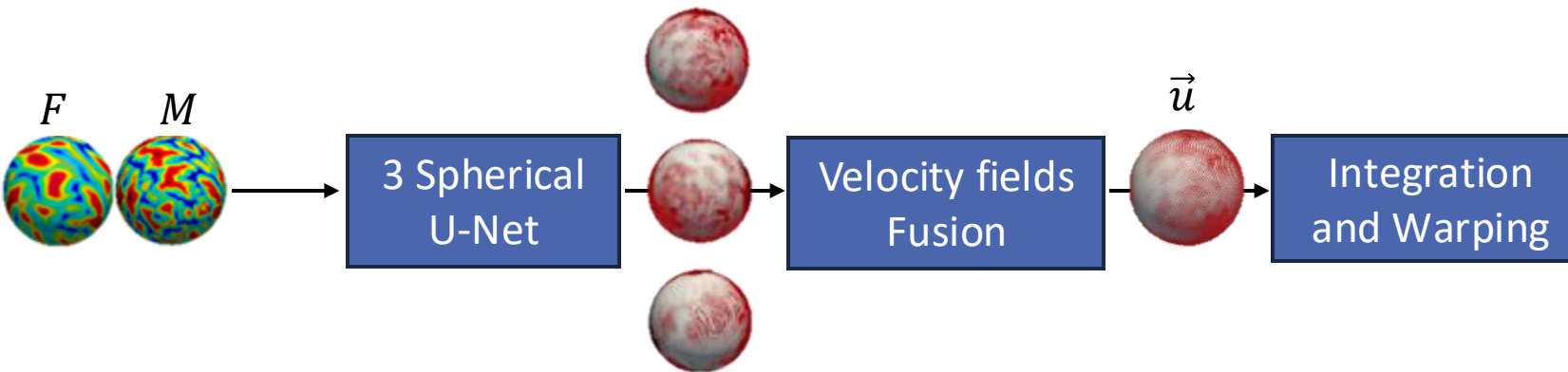
- Accurate alignment but slow
- Requires Multi-resolution scheme
- Remeshing to regular icosahedra as preprocessing

Icosahedron

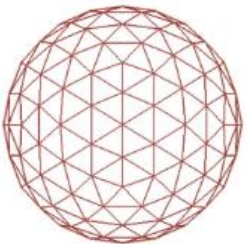


Spherical Surface Registration

S3Reg
(Zhao et al. 2021)



Icosahedron



⚠ Polar distortions → 3 orthogonal subnetworks

⚠ Initial displacement still needs to be small and smooth

✓ Faster computation times (s instead of min)

Spherical Surface Registration

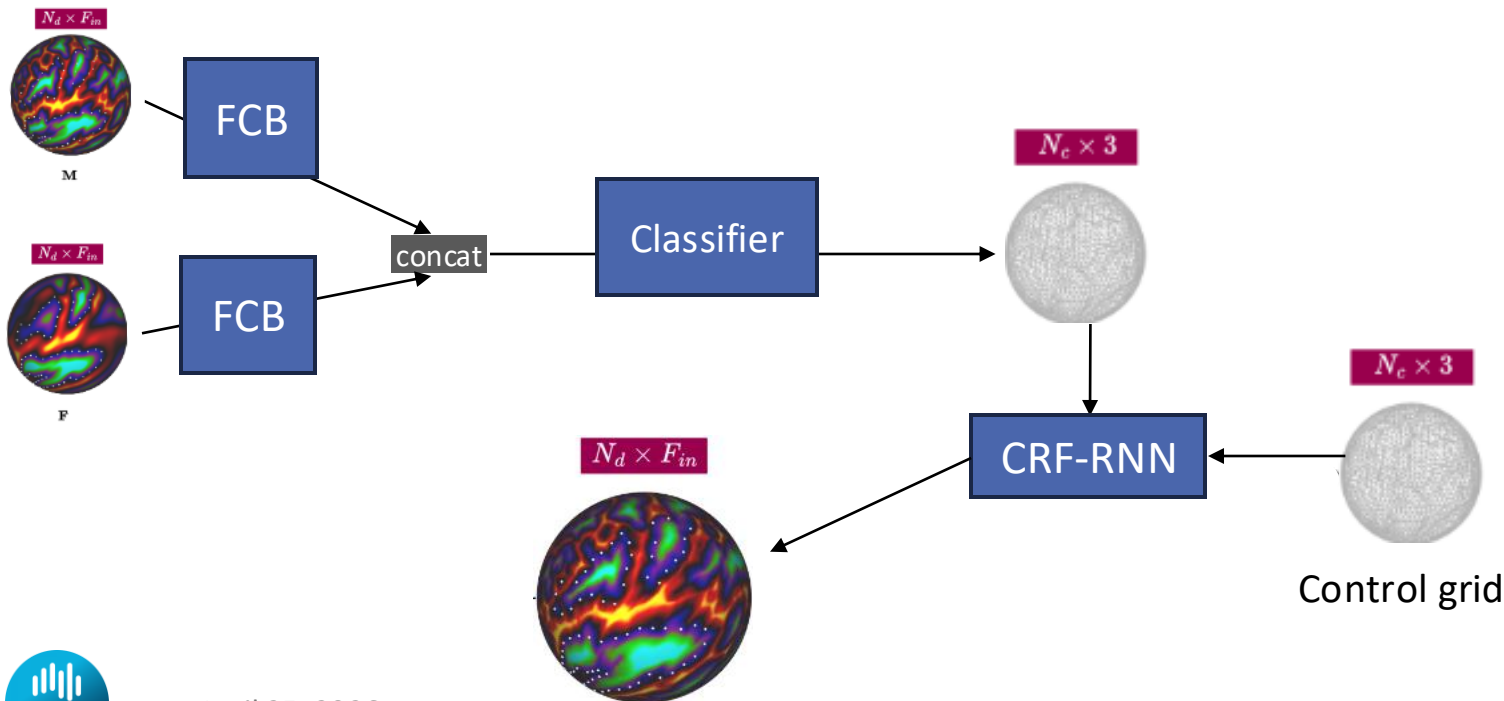
DDR and GeoMorph
(Suliman et al. 2022)

MoNet Convolution filter (Monti et al. 2017)

Classifier network : label $\leftrightarrow$ displacement

CRF-RNN imposes smoothness

+ Feature extraction network (FCB) +
additional regularization minimizing local
distortion during training



Deep Learning Methods

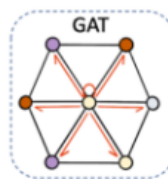
S3Reg
(Zhao et al. 2021)

Multiresolution approach

SUGAR (MB)	lh_Rigid_904_fsaverage6	223
	lh_NoRigid_904_fsaverage3	8.52
	lh_NoRigid_904_fsaverage4	26.51
	lh_NoRigid_904_fsaverage5	76.50
	lh_NoRigid_904_fsaverage6	223

Spherical Surface Registration

DDR and GeoMorph
(Suliman et al. 2022)



Spherical **graph attention** network (S-GAT)

Outputs local Euler angles

$$\mathcal{L} = \mathcal{L}_{sim} + \mathcal{L}_{Dice}$$

$$+ \mathcal{L}_{dist} + \mathcal{L}_{areal} \\ + \mathcal{L}_{angle} + \mathcal{L}_{folds}$$

Regularization

Deep Learning Methods

S3Reg
(Zhao et al. 2021)

DDR and GeoMorph
(Suliman et al. 2022)

Spherical Surface Registration

SUGAR
(Ren et al. 2024)

- ✓ Faster than traditional methods
- ⚠ Require substantial resources for training
- ⚠ Do not generalize well
- 🔨 Remeshing to regular icosahedra as preprocessing

Spherical Surface Registration - Summary

Non-Deep Learning Methods

SD

MSM

HSD

Deep Learning Methods

S3Reg

DDR

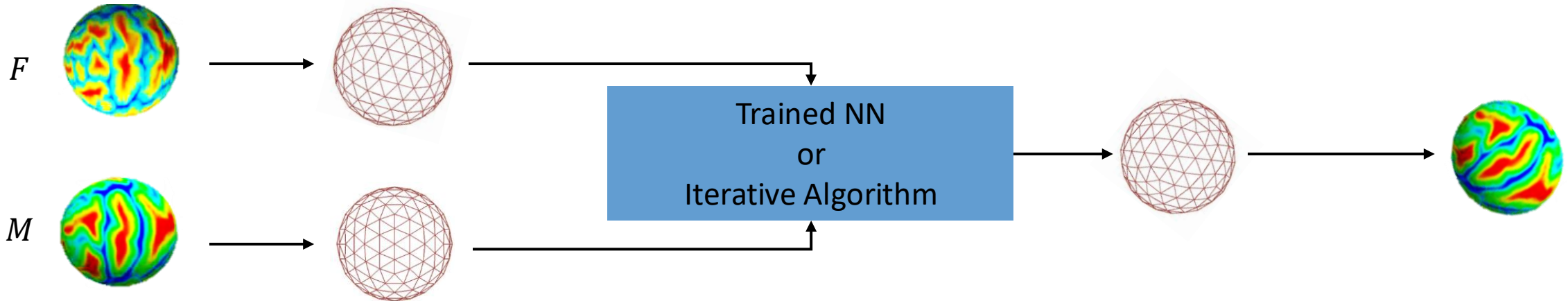
GeoMorph

SUGAR

Original **high-resolution** meshes
~ 100,000 vertices

Regular Icosahedron
< 10,242 vertices
(order 5)

Warp and Up-sample
to the **next resolution**



What are the **current**
strategies for rotational
pre-alignment ?



Spherical Pre-Alignment

Challenges of existing NRR

Tricky processing of meshes

Strong reliance on rigid
pre-alignment

non-Euclidean and non-convex

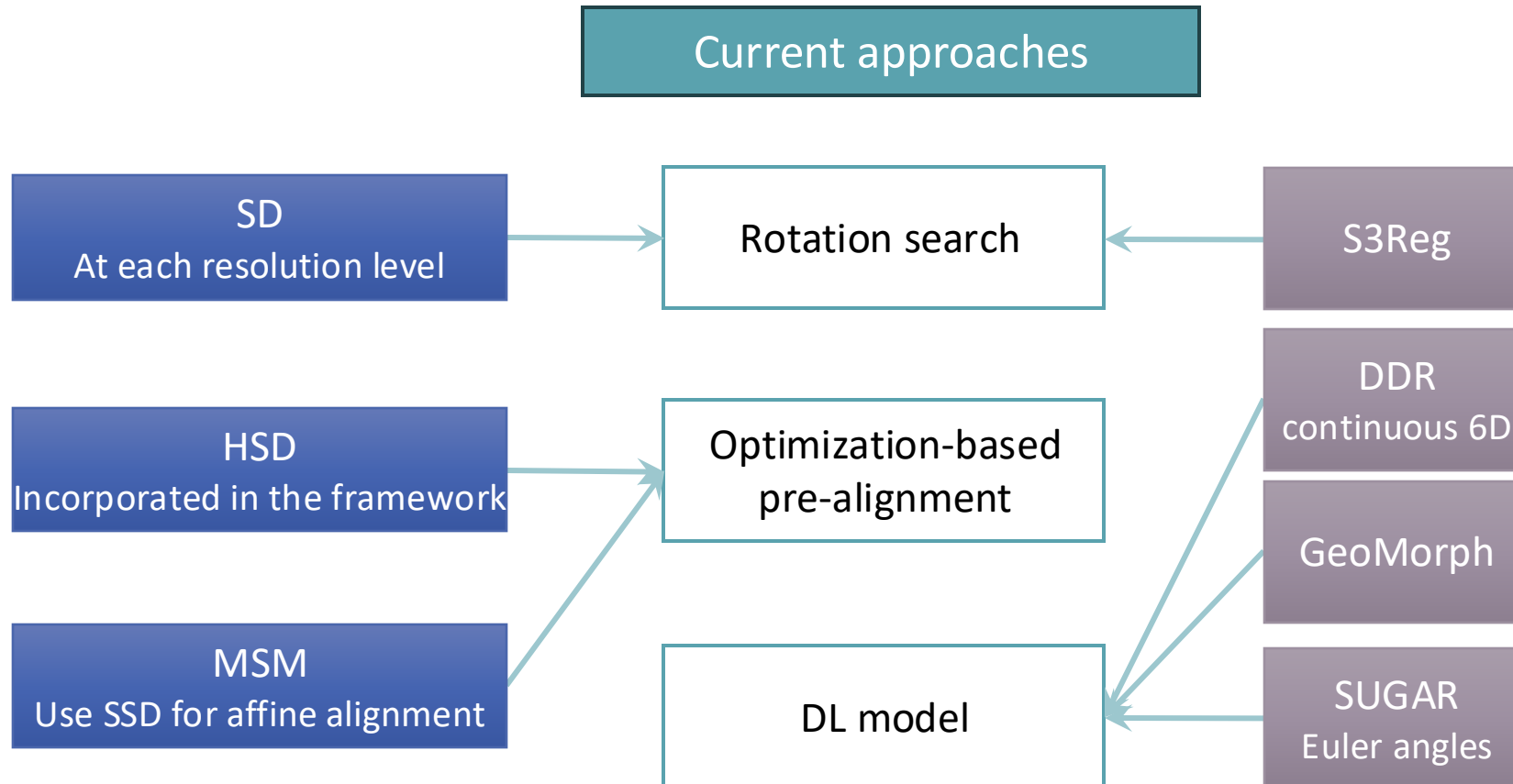
Many local minima exists

Problem formulation

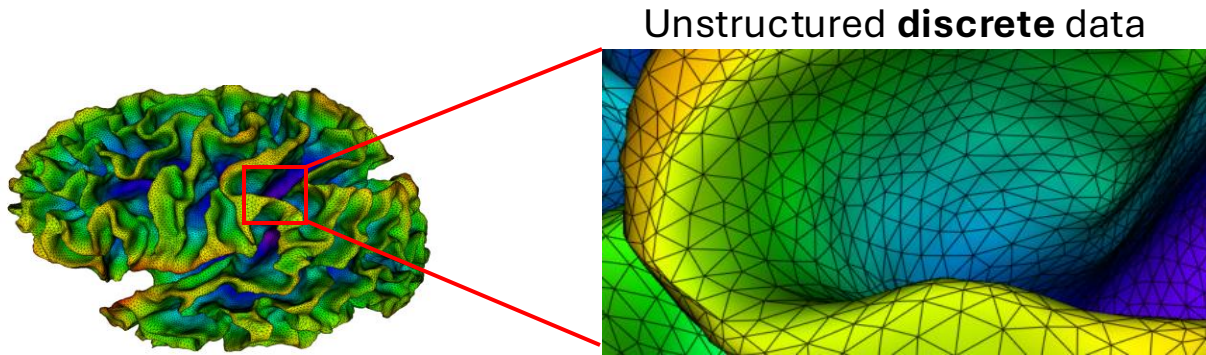
$$\min_{R \in SO(3)} \mathcal{L}(F, M, R)$$

$$\mathcal{L}(F, M, R) = \int \text{MSE}(F(p), M(R \cdot p)) dp$$

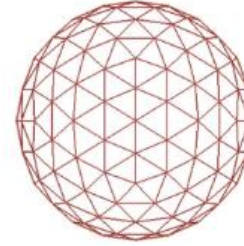
Spherical Pre-Alignment



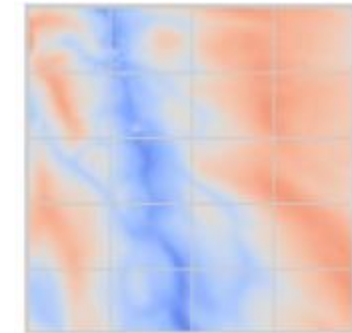
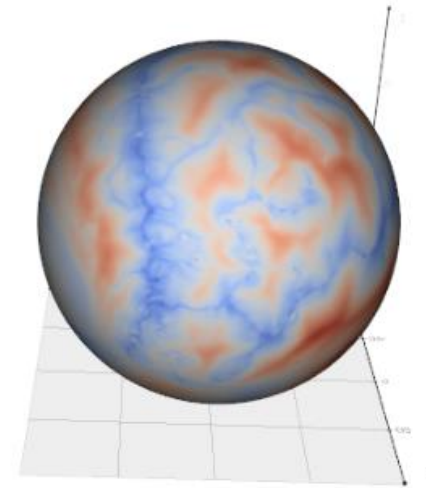
Operations on meshes are error-prone and slow



Icosahedron



"Icosahedrons can be efficiently **up and downsampled** to different resolutions"
(Fawaz et al. 2021)



Projection and 2D interpolation

Main Takeaways

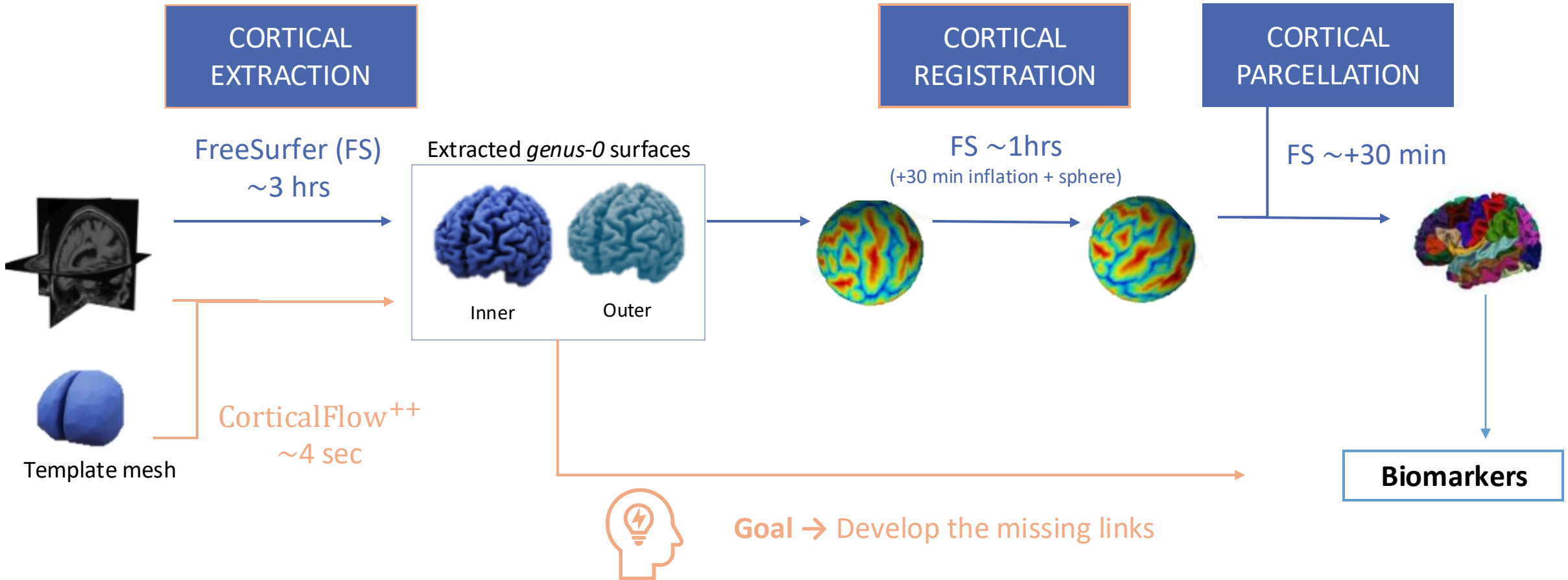
Registration methods

- ✓ Fast inference times for DL methods
- ✓ Spherical Demons: theoretical existence and regularization guarantees
- ⚠ Lack of generalization of DL methods

Neural Representation

- ✓ Alternative to tricky discrete representation
- ⚠ Absence of such a representation for the cortex in the literature

2D-surface based processing



Outline

Background

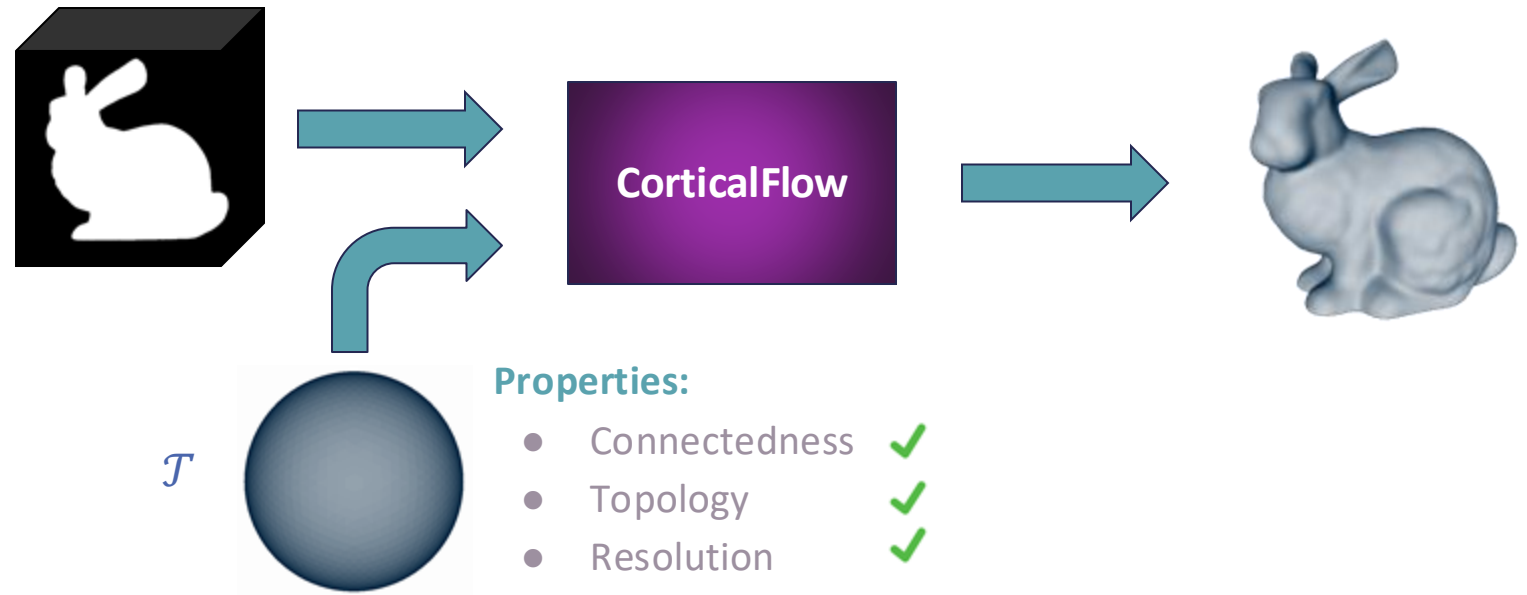
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Let

$$f_{\theta}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

Be a **diffeomorphism** and $\mathcal{T}$ a smooth closed 2-manifold of genus 0 embedded in $\mathbb{R}^3$, then the homotopy classes of $f_{\theta}(\mathcal{T})$ and $\mathcal{T}$ are the same.

$\approx$ we cannot introduce **holes** or **self-intersections** in the deformed template.

Diffeomorphic Mesh Deformation (DMD)

Solving flow ODE

$$\frac{d\phi(t;x)}{dt} = U(\phi(t;x)), \text{ with } \phi(0;x) = x$$

Forward Euler method

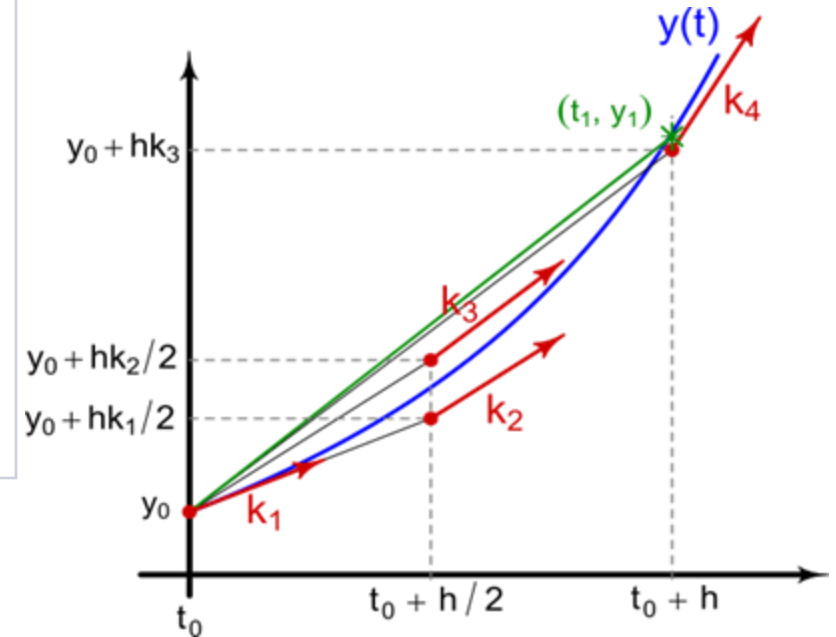
$$\hat{\phi}(h, x) = x + hU(x)$$

CorticalFlow

RK4 method

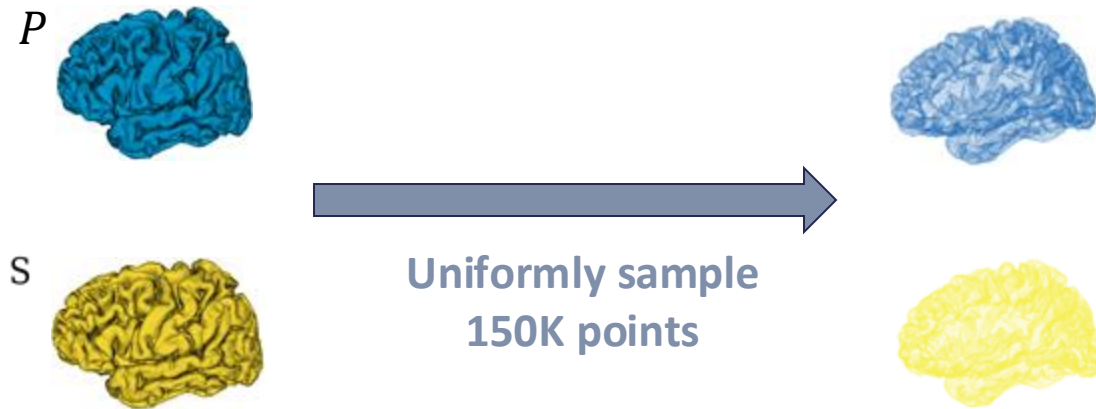
$$\hat{\phi}(h, x) = x + \frac{h}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

CorticalFlow⁺⁺



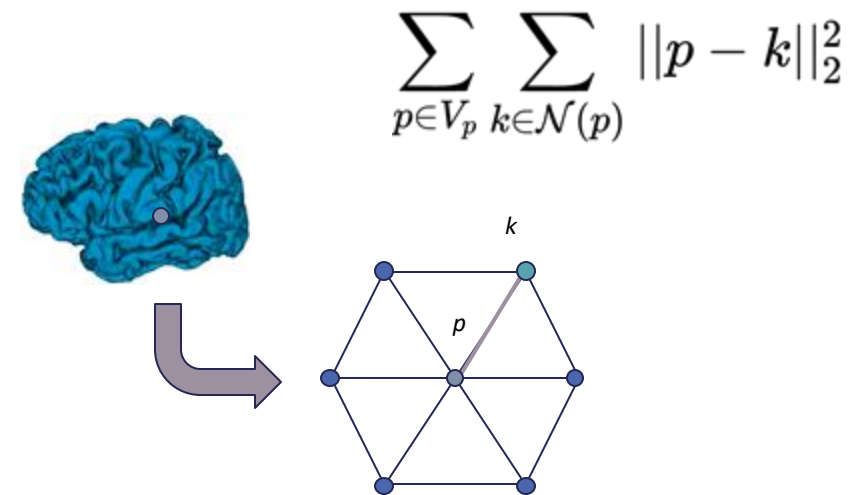
Loss function

Chamfer distance



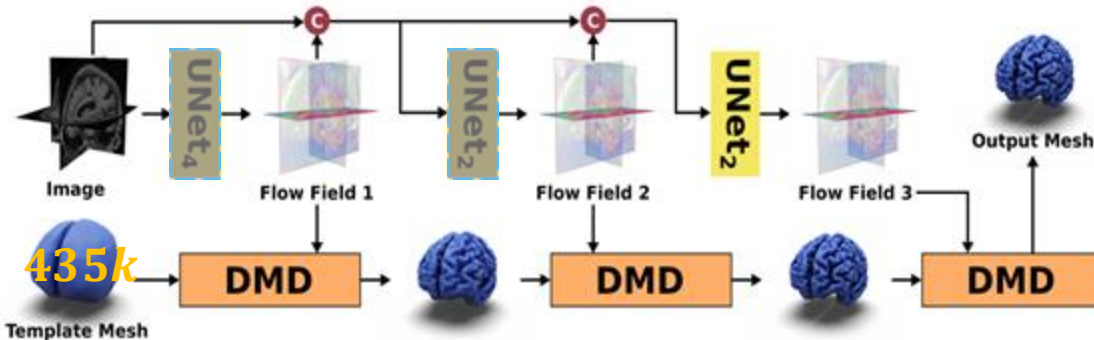
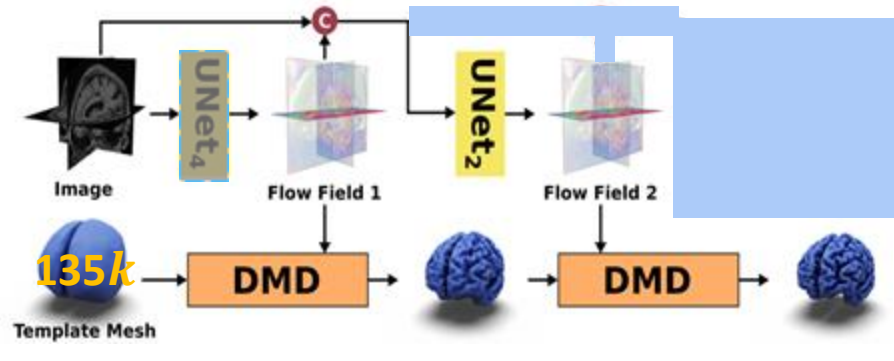
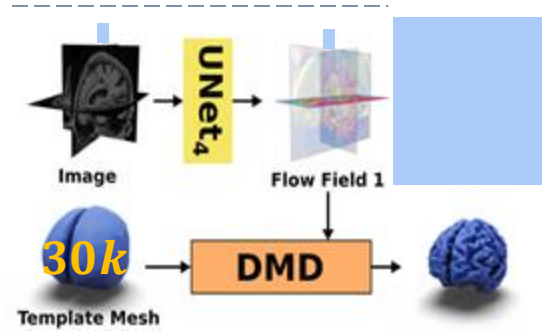
$$\frac{1}{N} \left(\sum_{p \in P} \min_{s \in S} \|p - s\|_2^2 + \sum_{s \in S} \min_{p \in P} \|p - s\|_2^2 \right)$$

Edge length regularizer



$$\operatorname{argmin}_{\theta_i} \sum_{(I, S) \in \mathcal{D}} \mathcal{L} \left(CF_{\theta_i}^i(I, \mathcal{T}_i), S \right)$$

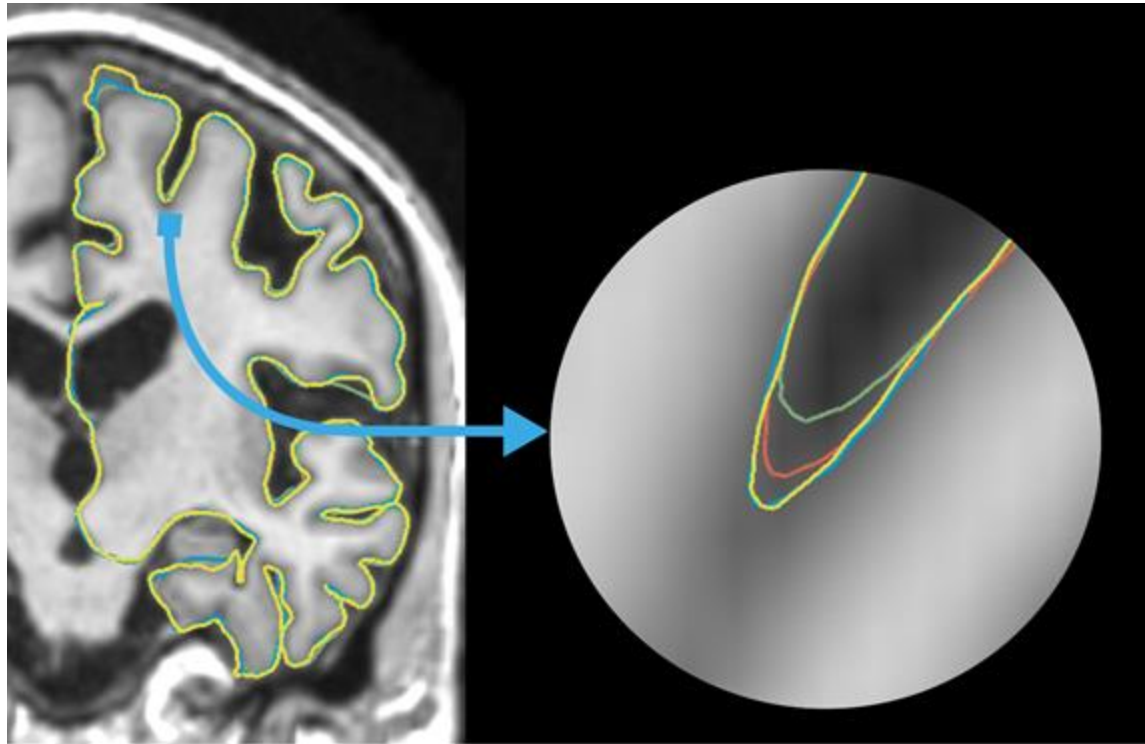
Multiscale training



Cascaded UNets

- Sequential training
- At each step, the new network takes as input the images and the preceding flows
- At each step, the mesh resolution is increased (from 30k to 435k)

Multiscale training

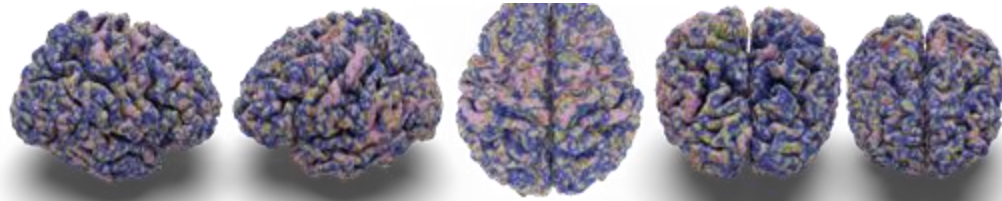


— 1st Deformation — 2nd Deformation
— 3rd Deformation — pseudo-ground-truth

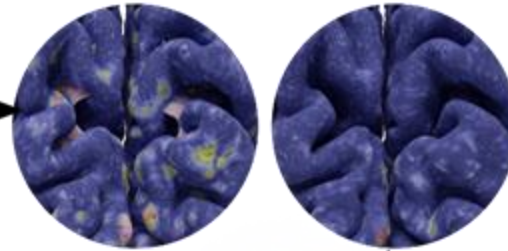
Allows the reconstruction of **more accurate** surfaces.

Geometric Accuracy

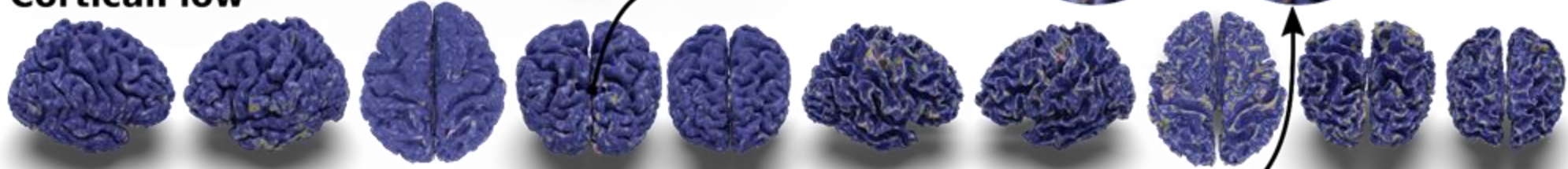
PiaINN[★] (Ma et al., 2021)



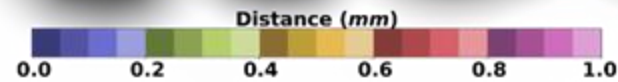
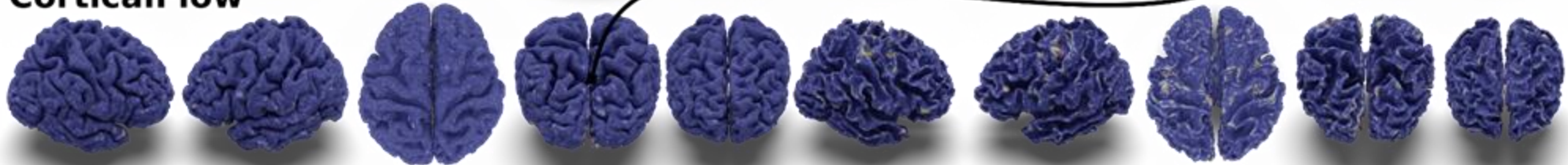
Reduced Mesh Artifacts



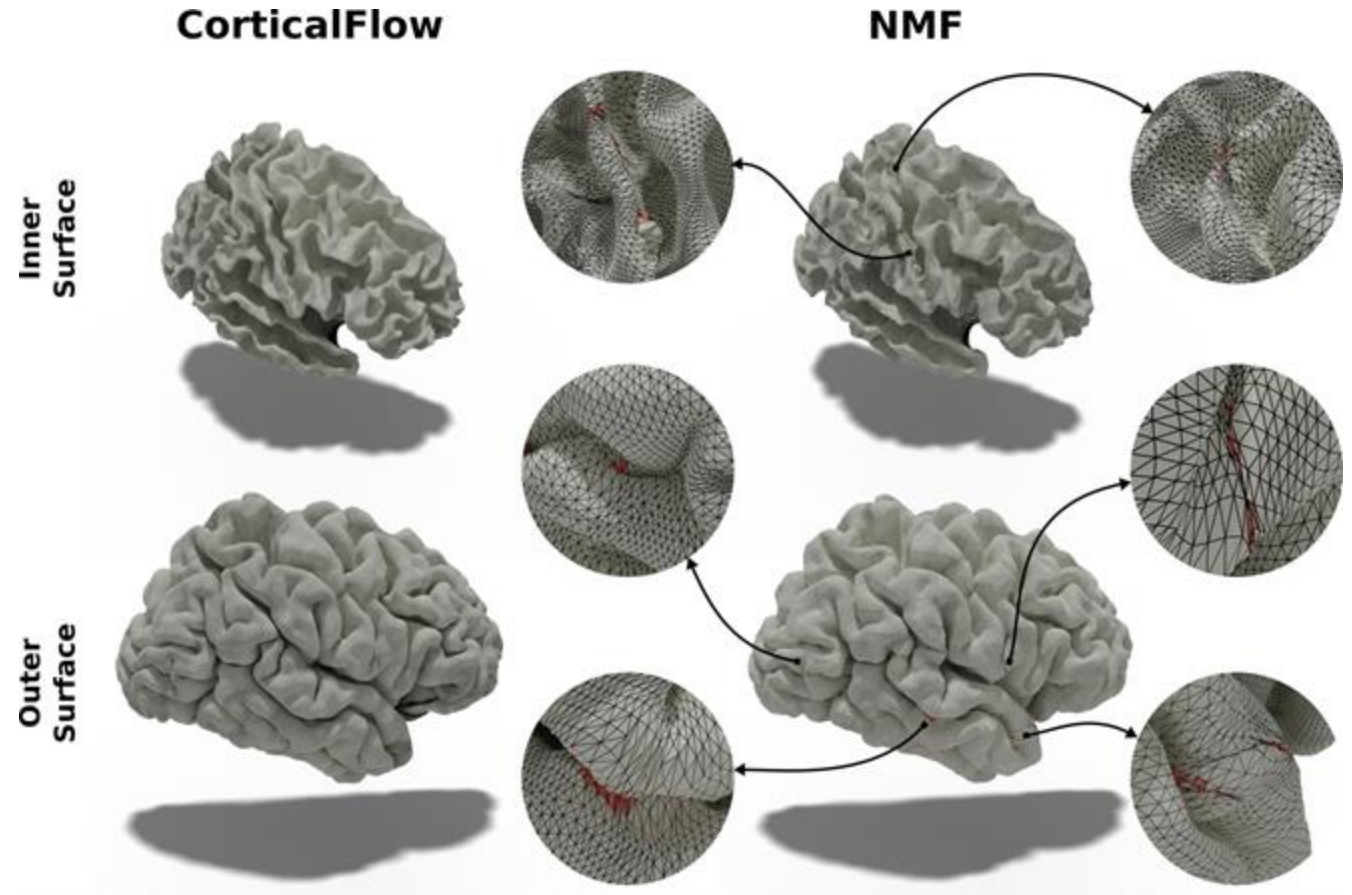
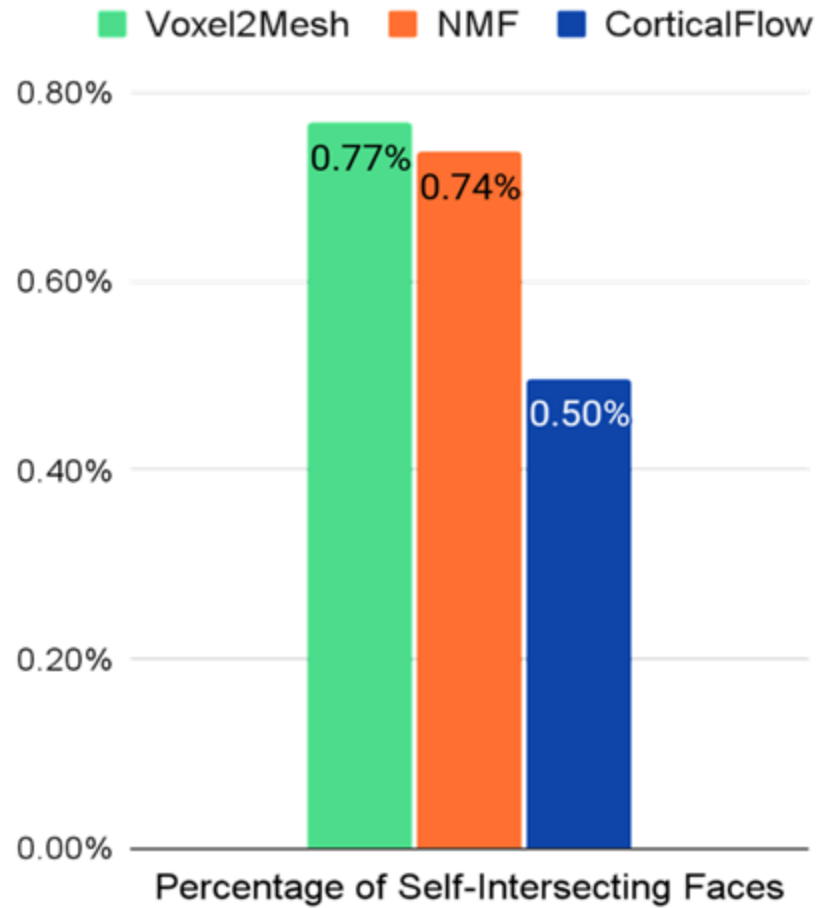
CorticalFlow



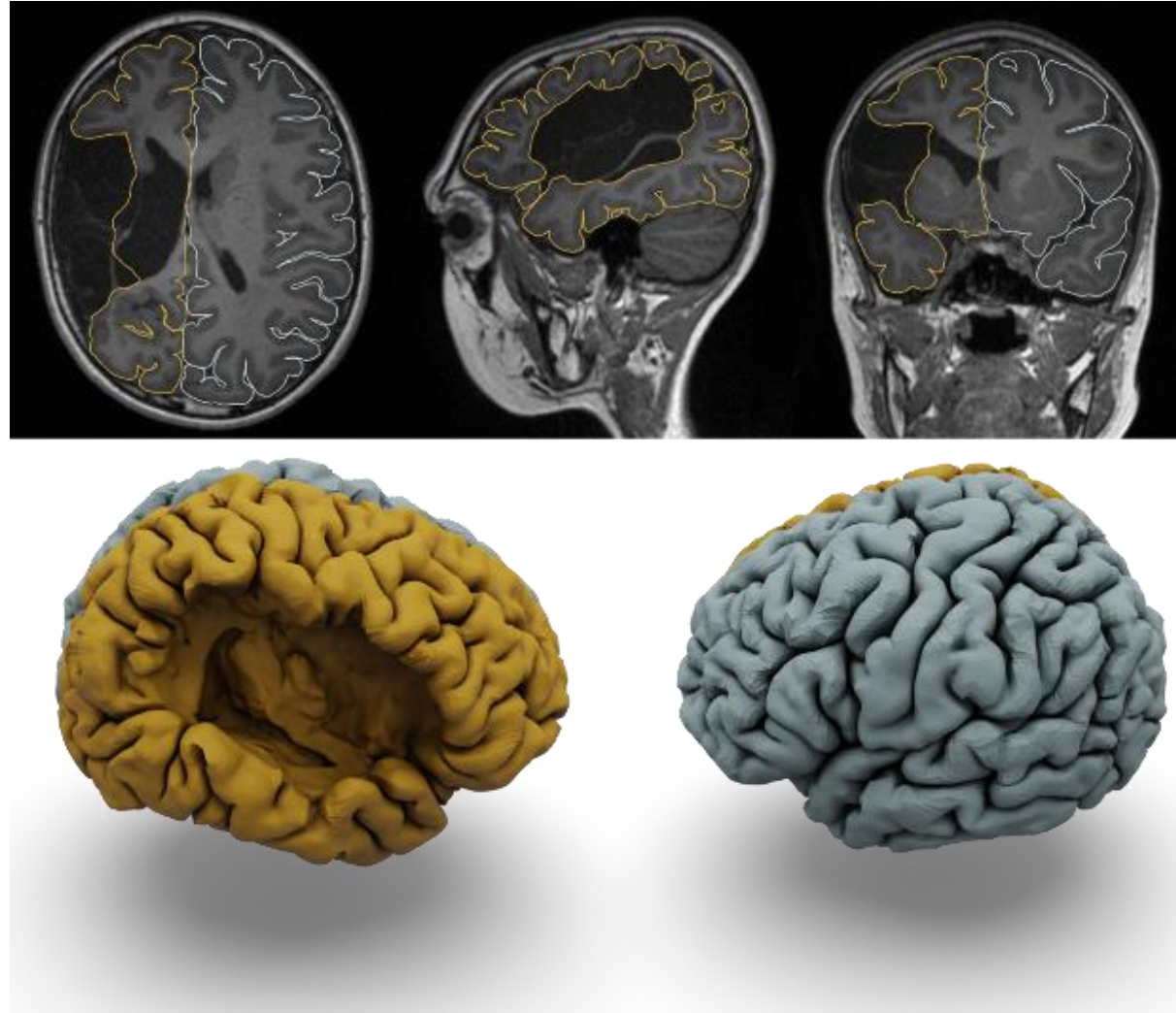
CorticalFlow⁺⁺



Surface Regularity



Testing on Challenging Anatomies (Cerebral Palsy)



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Understanding Spherical Demons

- Reminder of the [Registration problem formulation](#)
- For regularization: Hidden variable $\gamma \rightarrow$ Two step optimization

$$\arg \min_{\Gamma, \gamma} \underbrace{\frac{1}{\sigma_i^2} ||F - M \circ \Gamma||^2 + \frac{1}{\sigma_x^2} \text{dist}(\Gamma, \gamma)^2}_{\text{Step 1: Optimize } \Gamma} + \underbrace{\frac{1}{\sigma_T^2} \mathcal{R}(\gamma)}_{\text{Step 2: Optimize } \gamma}$$

Understanding Spherical Demons

1. Write Γ as the integration of a SVF: $\Gamma = \exp(v)$

→ Optimisation over v **Step 1: Optimize Γ**

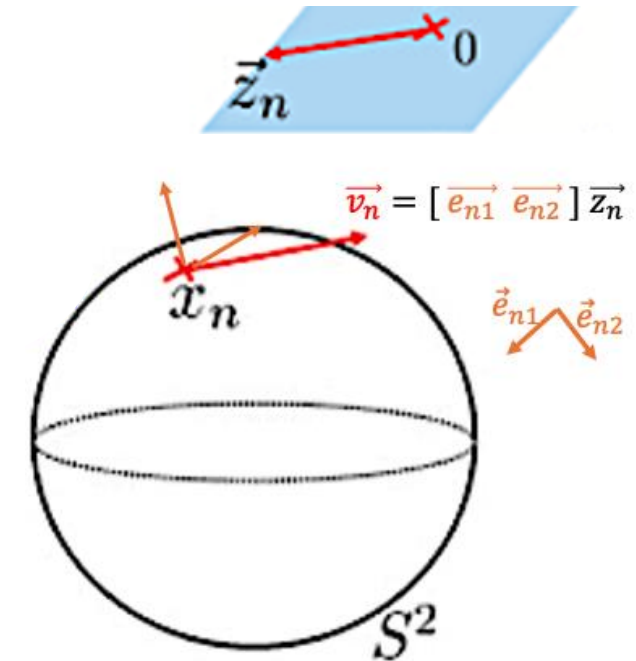
2. Write v as a tangent vector field $\vec{v}_n \in T_{x_n} S^2$

→ Constrained optimization
$$\arg \min_{\Gamma} \frac{1}{\sigma_i^2} \|\Gamma - M \circ \Gamma\|^2 + \frac{1}{\sigma_x^2} \text{dist}(\Gamma, \gamma)^2$$

3. Write $\vec{v}_n$ using local chart coordinates

→ Unconstrained optimization over $\vec{z}_n \in \mathbb{R}^2$

→ Existence of a solution and Gauss-Newton optimization



Understanding Spherical Demons

Step 2: Optimize γ

$$\arg \min_{\gamma} \frac{1}{\sigma_x^2} \text{dist}(\Gamma, \gamma)^2 + \frac{1}{\sigma_T^2} \mathcal{R}(\gamma)$$

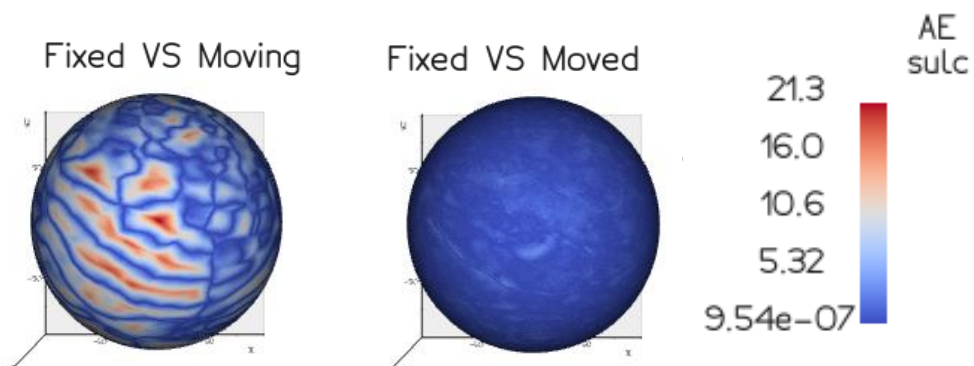
- Inexact Vector Spline Interpolation Problem (Glaunès et al. 2004)
 - The chosen regularization operator is related to a Reproducing Kernel Hilbert Space (RKHS)
- **Uniqueness** and **Existence** of a solution depending on the **reproducing kernel** K
 - ⚠ Computation of a Large size reproducing kernel matrix
- Approximate explicit matrix with an approximation of K
 - Approximation by Iterative smoothing

$$\hat{\gamma} = K^{\text{approx}} \vec{\Gamma}$$

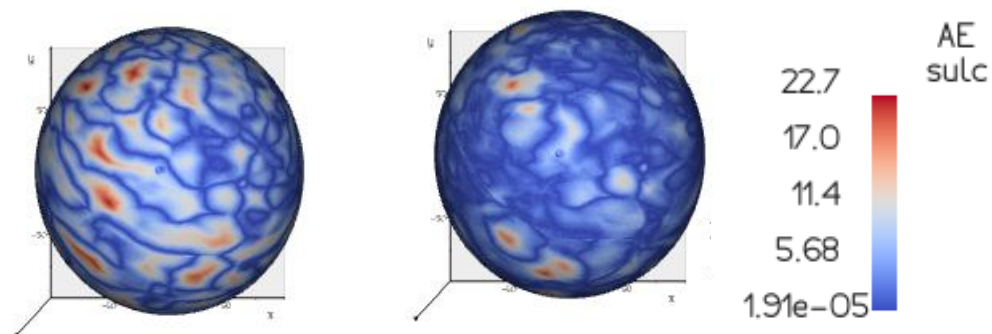
Understanding Spherical Demons



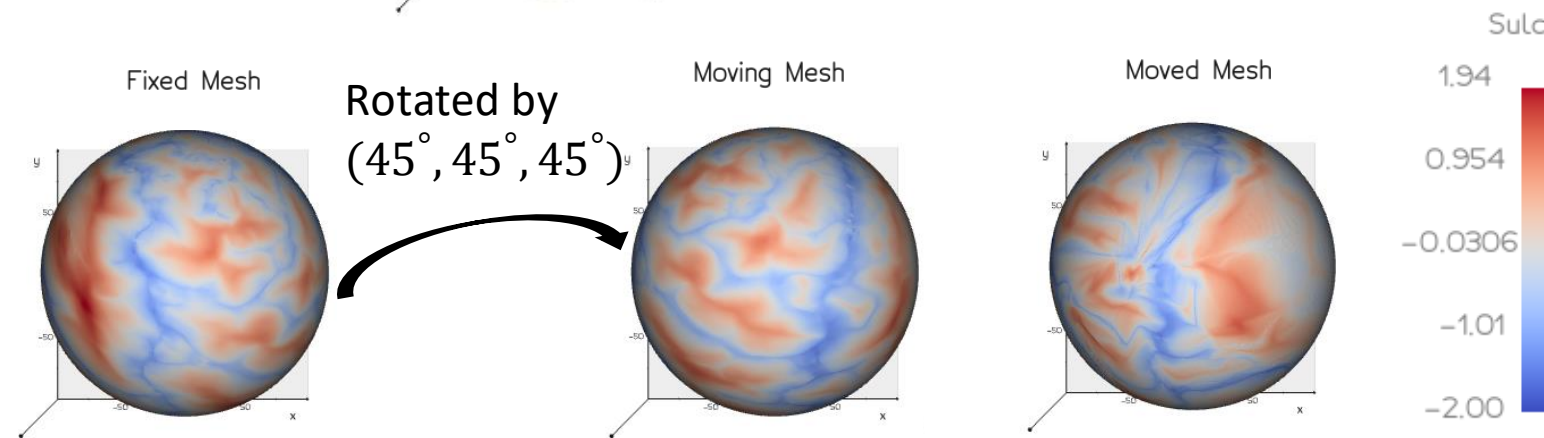
Same subject, different time points



Different subjects



No pre-alignment



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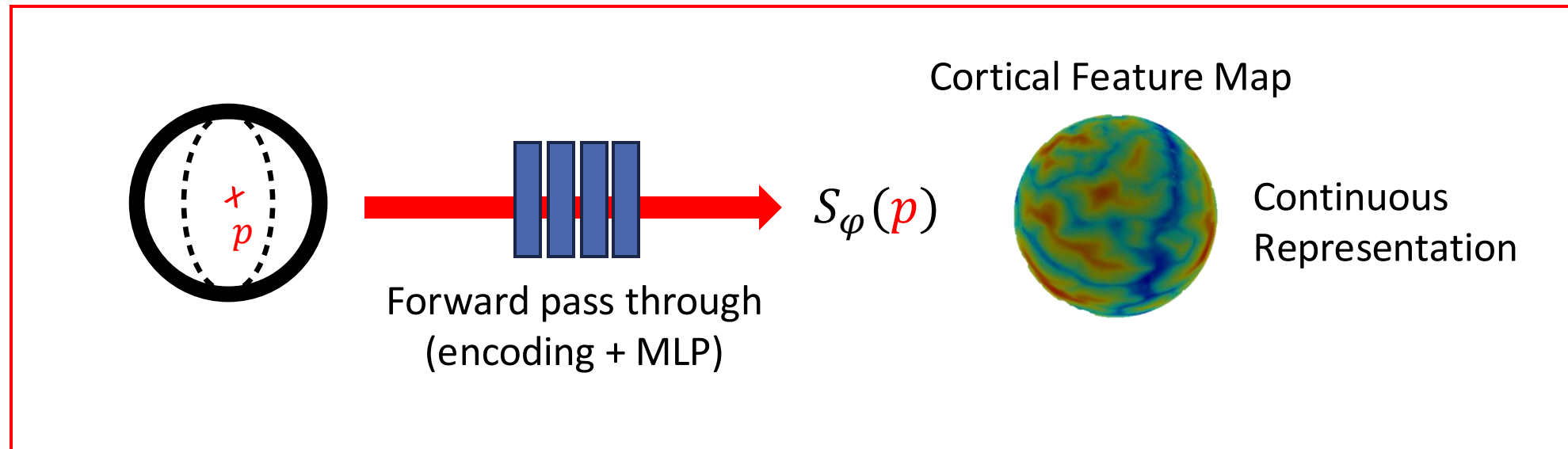
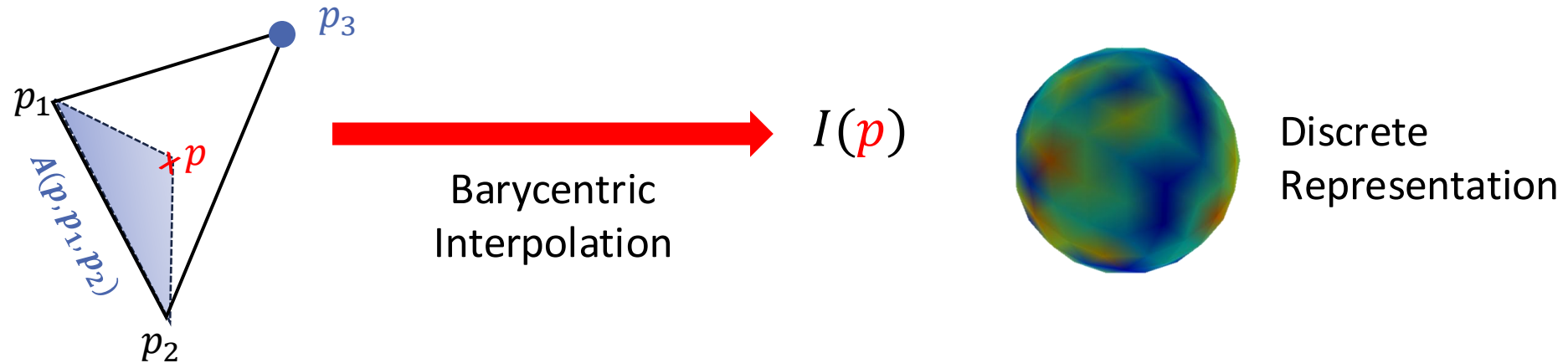
Conclusion

Our Goals

1. Propose a continuous representation
2. Propose a registration strategy demonstrating the potential of this representation



Neural Cortical (NC) Maps



Neural Cortical Maps (NC) - Definition

Model

Encoding + Fully-fused
MLP from [tiny-cuda-nn](#)

$$\mathcal{S}_\varphi: \mathbb{S}^2 \rightarrow \mathbb{R}^{n_f}$$

Training

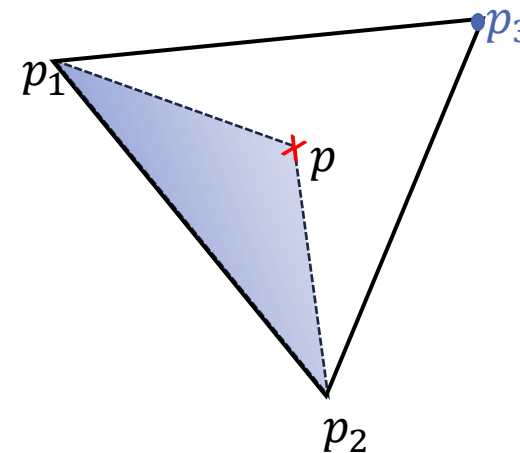
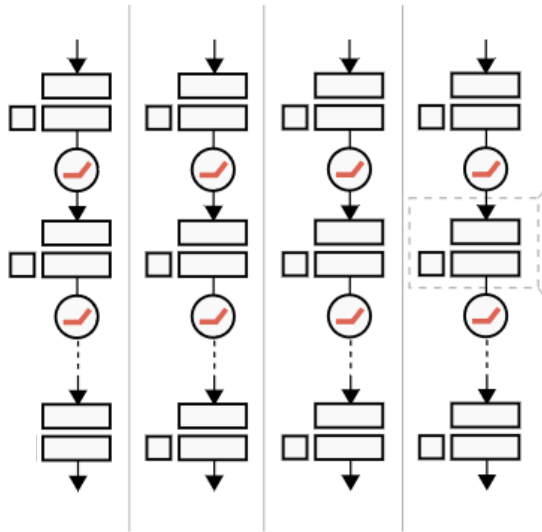
$$\arg \min_{\varphi} \int_{p \in \mathbb{S}^2} \mathcal{L}(\mathcal{S}_\varphi(p), I(p)) dp$$

Sampling faces

Sampling random
points on the face

Learn interpolation

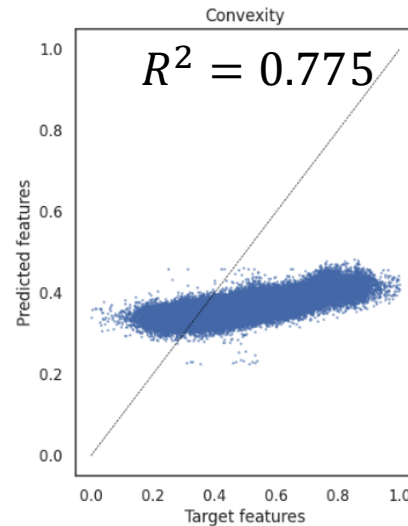
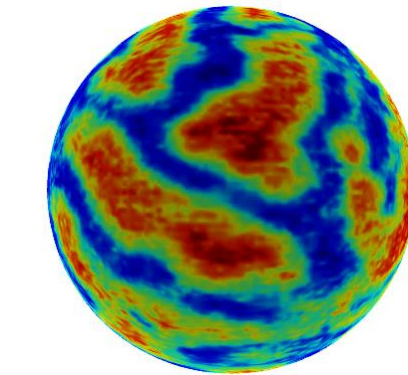
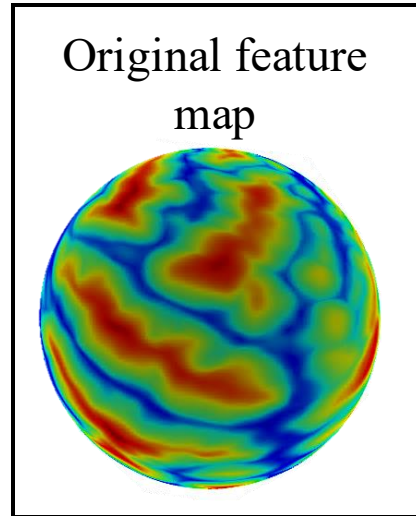
(Müller et al. 2021)



How well NC maps **learn**
features ?

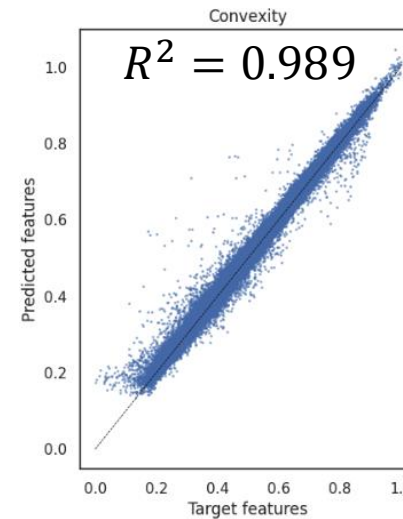
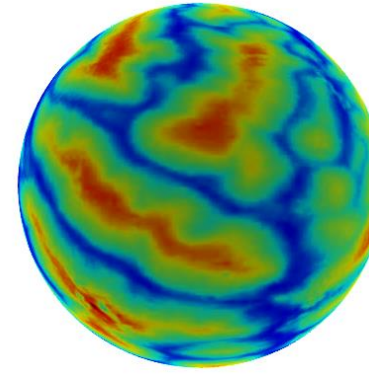


Neural Cortical Maps (NC) - Experiments



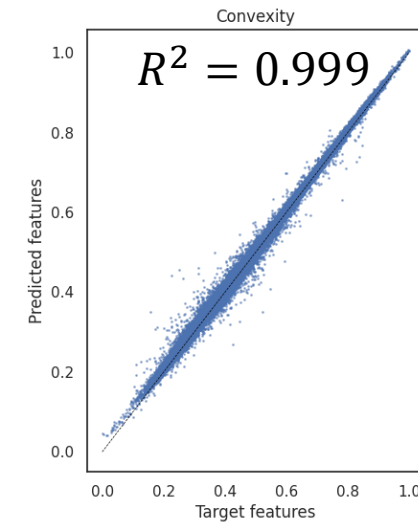
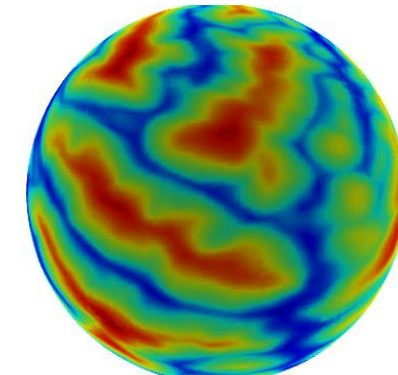
$N_{\text{iter}} = 10$

0.7s



$N_{\text{iter}} = 100$

0.8s



$N_{\text{iter}} = 1000$

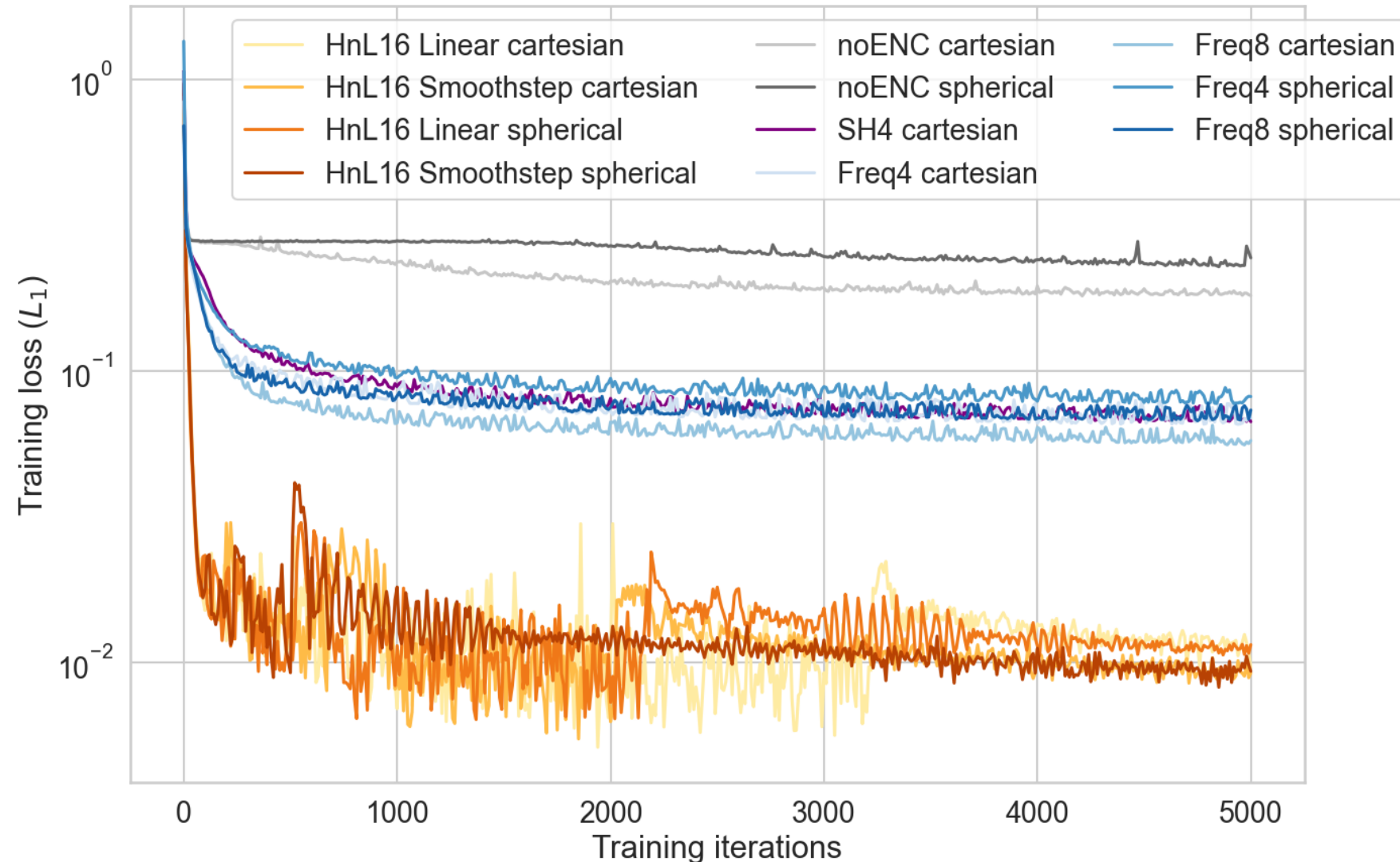
6.6s

Time

How does input **encoding**
influence **performance** ?

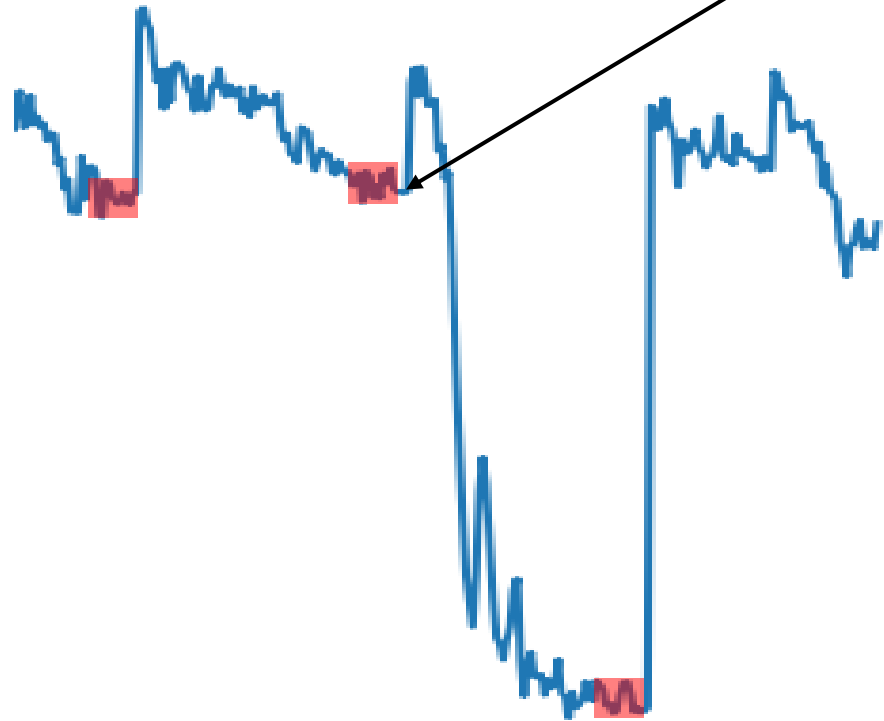


Neural Cortical Maps (NC) - Experiments



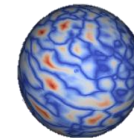
NC-Reg with Simulated Annealing (SA)

$$\Delta\mathcal{L} < 0 \quad \text{OR} \quad \mathcal{U}([0,1]) < \exp\left(\frac{-\Delta\mathcal{L}}{T_t}\right)$$

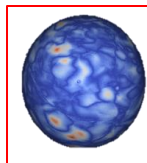


Local minimum

$$\Delta\mathcal{L} = \mathcal{L}(F, M, R_{rand}) - \mathcal{L}_t$$



Feature difference



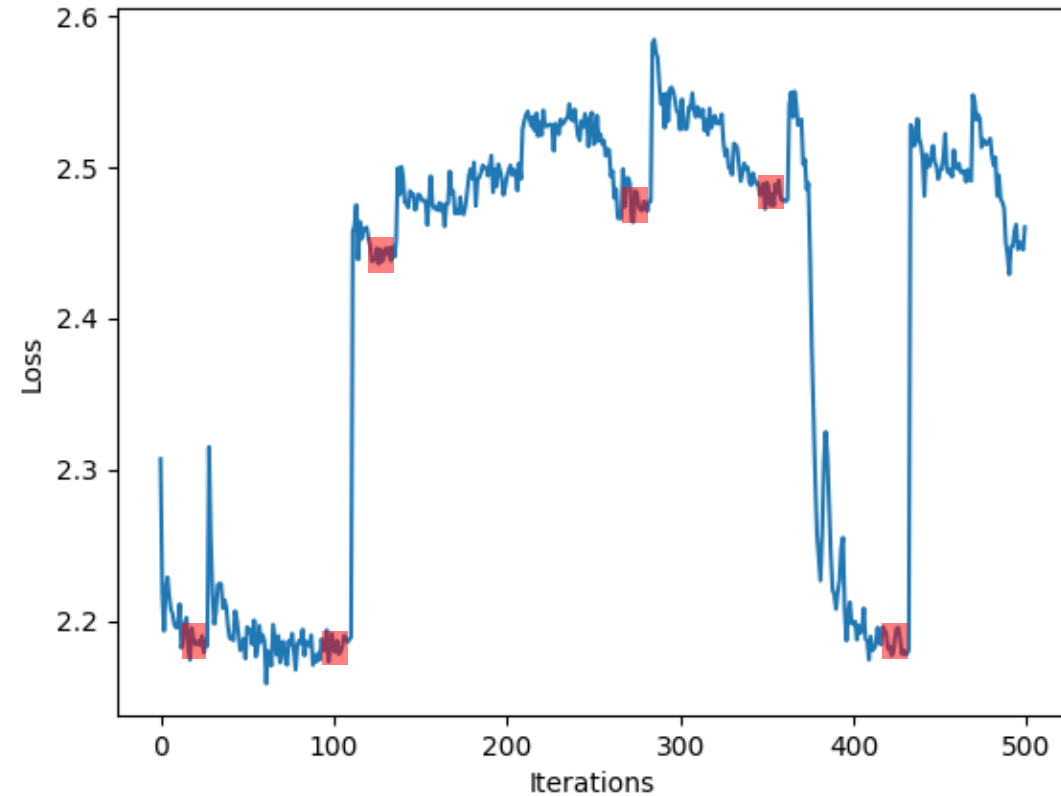
How to identify local minima ?

Moving average

$$\bar{\mathcal{L}}_t = \sum_{i=t-t_m}^{t-1} \mathcal{L}_i$$

Local
minima ?

$|\mathcal{L}_t - \bar{\mathcal{L}}_t| < \epsilon$
for n
subsequent
iterations



Experimental Setup

Task: Register

Template *NC map*
Trained once on
fsaverage6

to

Dataset

ADRC
39 subjects with
manual
parcellations

Perturbed
ADRC
Rotated by
 $\mathcal{U}(-36^\circ, 36^\circ)$

Robustness to
initialisation

Baselines

SD

(Yeo et al., 2010)

HSD

(Lyu et al., 2019)

SUGAR

(Ren et al., 2024)

Evaluation metrics

MSE, PCC

Feature
alignment

Dice Score

Brain region
overlap

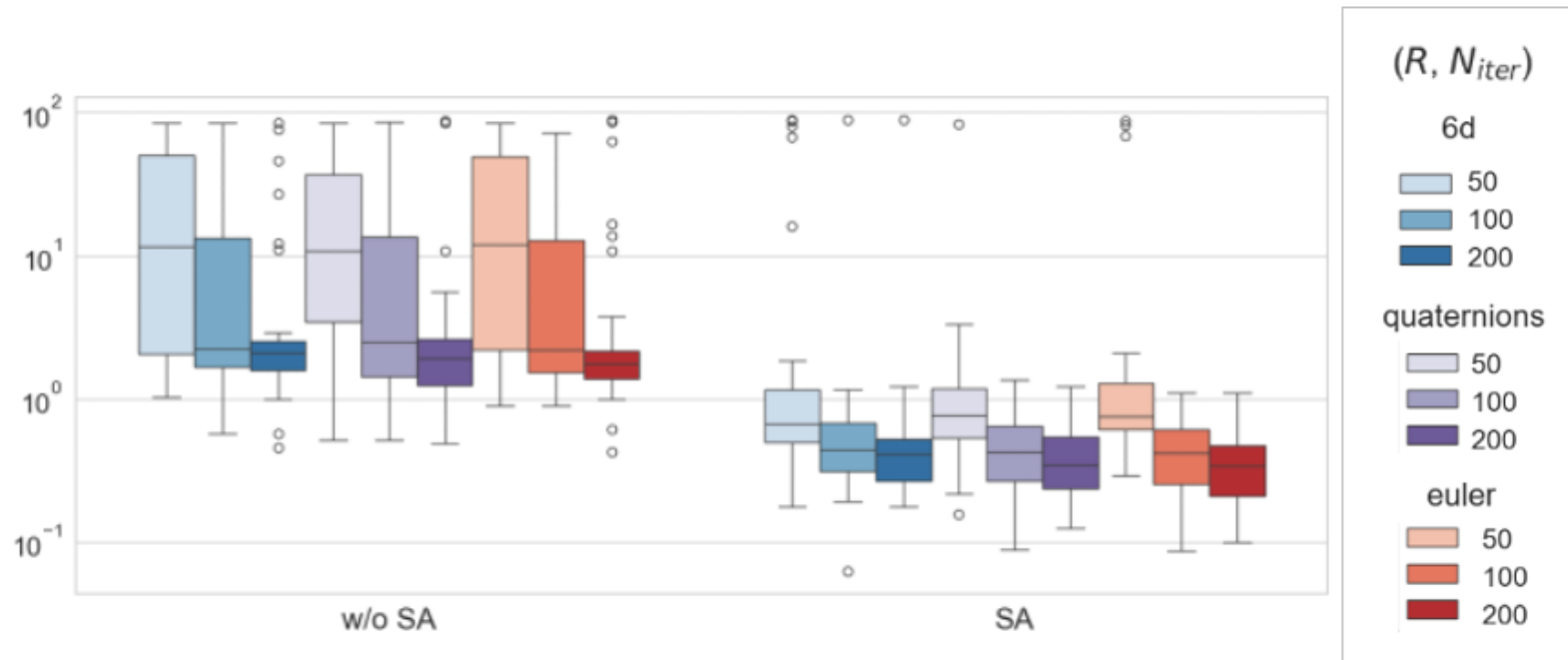
Quaternion
error

Compare
rotations

NC-Reg : Relevance of SA

Same **solution** for **different** initialisation ?

↓ Quaternions
error (°)
(Huynh, 2009)

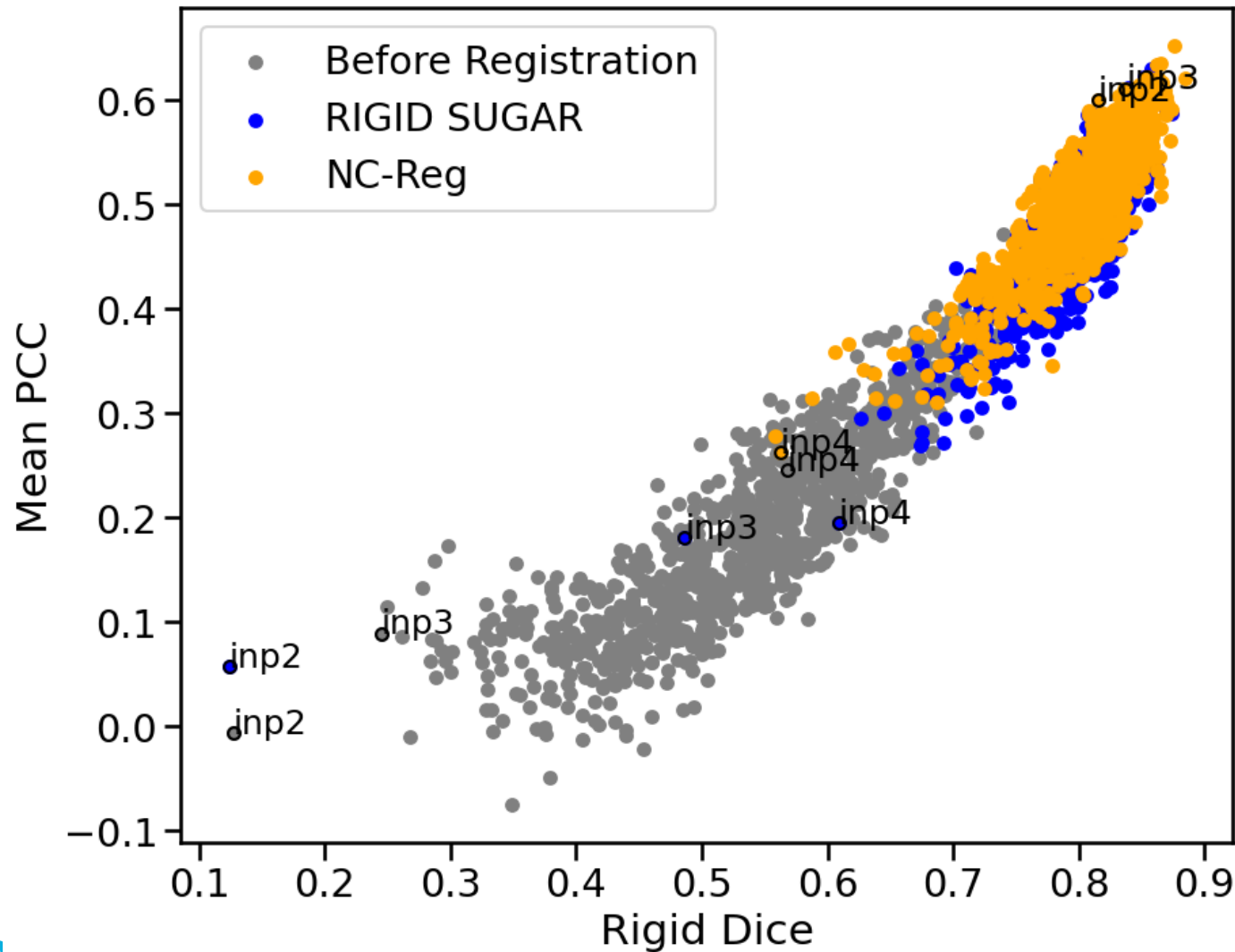


NC-Reg : Integration in NRR workflows

NRR on the *perturbed* dataset

Method	MSE (↓)	PCC (↑)	Dice (↑)	Time (s)	Worst Dice
	sulc	sulc			
SD	0.057	0.912	0.812	35.55	0.0%
NC-Reg + SD	0.056	0.914	0.821	30.26	0.0%
SUGAR	0.167	0.731	0.412	1.46	7.7%
NC-Reg + SUGAR	0.063	0.903	0.822	2.13	0.0%

Key Takeaways



 Robust, Consistent and Accurate

 Runtimes

 More hyperparameters but

$$100 \leq N_{\text{iter}} \leq 500$$

$$\epsilon \leq 0.05$$

$$T_0 \leq 0.5$$

$$T_{\text{min}} = 1e^{-4}$$



 Exploring other variants of SA
Exploring faster way to identify local minima

Outline

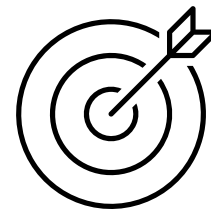
Background

Literature Review

Presentation of CorticalFlow⁺⁺

Insights into Spherical Demons

Presentation of NC-Reg



Conclusion

Conclusion



NC maps

- Novel continuous representation
- Small MLP, fast to train
- Promising potential for classical tasks
- Application to Rigid registration with NC-Reg



Integrate NC maps into an end-to-end neuroimaging pipeline

Thank You !

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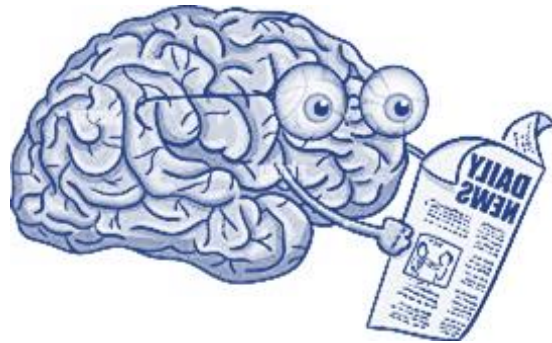
[inesvati.github.io](https://github.com/inesvati)



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Appendix



Diffeomorphic Mesh Deformation (DMD)

The deformation field is

$$\phi : (t, x) \in [0, 1] \times \mathbb{R}^3 \rightarrow \phi(t, x) \in \mathbb{R}^3$$

$\mathcal{V}$ being the set of vertices, $U : \mathbb{R}^{|\mathcal{V}| \times 3} \rightarrow \mathbb{R}^{|\mathcal{V}| \times 3}$

Forward Euler method

$$y_{k+1} = \hat{\phi}(t_{k+1}, x) = \hat{\phi}(t_k, x) + hU\left(\hat{\phi}(t_k, x)\right), \text{ with } t_k = kh \text{ and } h = 1/K$$

RK4 method

$$\hat{\phi}(t_{k+1}, x) = \hat{\phi}(t_k, x) + \frac{h}{6}[k_1 + 2k_2 + 2k_3 + k_4]$$

$$k_1 = U(x)$$

$$k_2 = U\left(x + \frac{h}{2}k_1\right)$$

$$k_3 = U\left(x + \frac{h}{2}k_2\right)$$

$$k_4 = U(x + hk_3)$$

MoNet Convolution Filter

$x \in \mathbb{R}^d$: Point on a manifold.

$y \in \mathcal{N}(x)$: point in the neighborhood of x .

Pseudo coordinates $u(x, y) \in \mathbb{R}^d$

A learnable weighting function (kernel) $w_\Theta(u) = (w_1(u), \dots, w_J(u))$. J : dimension of the extracted patch

MoNet (Mixture Model Network) Convolution on non-Euclidean domains

$$(f \star g)(x) = \sum_{j=1}^J g_j D_j(x) f$$

They show that other methods (GCN, Diffusion CNN, Anisotropic CNN, Geodesic CNN) can be rewritten using particular definition of u and $w(u)$. In Monti et al

$$w_j(u) = \exp\left(-\frac{1}{2}(u - \mu_j)^T \Sigma_j^{-1}(u - \mu_j)\right)$$

where Σ_j and μ_j are learnable $d \times d$ and $d \times 1$ covariance matrix and mean vector of a Gaussian Kernel.

Patch operator

$$D_j(x)f = \sum_{y \in \mathcal{N}(x)} w_j(u(x, y))f(y)$$